Chapter 3

Surds and Indices

Index: $a \times a \times a \times a \times a$ .... m times

or $(a \times a \times a \times$ ....m times$) \times (a \times a \times a \times$ ....n times$)$

i.e. $a \times a \times a \times$ .... $(m + n)$ times

Important formulae : If $a > 0,\ a \neq 1,$ m and n are integers then

(i) $a^m \times a^n = a^{m+n}$ (ii) $a^m \times a^n \times a^Q = a^{m+n+Q}$ (iii) $(a^m)^n = a^{mn}$ (iv) $\frac{a^m}{a^n} = a^{m-n}$

(v) $a^0 = 1$ (vi) $a^{-m} = \frac{1}{a^m}$ (vii) $a^{m^n} = a^{(m)^n}$ (viii) $(ab)^n = a^n b^n$

Surd : If 'a' is a rational number and n is a positive integer such the n^{th} root of 'a', i.e., $a^{1/n}$ $\sqrt[n]{a}$ is an irrational number, then $a^{1/n}$ is called a surd. In other words, an irrational root of a rational number is called a surd.

Example: $\sqrt{2},\ \sqrt[3]{4},\ \sqrt[4]{18},\ \sqrt[7]{4},\ \sqrt[3]{9}$ etc. are surds

i.e. we can say that every number expressed in a surd is an irrational number.

Types of surds:

(i) Pure Surd: $\sqrt{7},\ 3\sqrt{11},\ \sqrt[4]{125}$ are pure surds.

(ii) Mixed Surd: $3\sqrt{2},\ 7\sqrt[2]{11},\ \sqrt{32}$ are mixed surds.

(iii) Similar surds: $3\sqrt{3},\ \sqrt{3}$ and $6\sqrt{5},\ 7\sqrt{125} = 35\sqrt{5}$

order of surds: $\sqrt{7},\ \sqrt[3]{4},\ \sqrt[4]{8}, \sqrt[5]{125}$ are respectively surds of order 2, 3, 4 and 5

Conjugate of surds : Two binomial surds which differ only in sign (+ or –) between the terms connecting them, are known as conjugate surds

Example : Conjugate of $5 + \sqrt{7}$ is $5 - \sqrt{7}$

Condition for two Surds to be equal : If a, b, c, d are all rational numbers and b and d are not perfect square then

$a + \sqrt{b} = c + \sqrt{d}$, i.e. a = c and b = d

Square root of surd of $a + \sqrt{b}$ form

$$\sqrt{a+\sqrt{b}} = \sqrt{\frac{a+\sqrt{a^2-b}}{2}} + \sqrt{\frac{a-\sqrt{a^2-b}}{2}}, \qquad \sqrt{a-\sqrt{b}} = \sqrt{\frac{a+\sqrt{a^2-b}}{2}} - \sqrt{\frac{a-\sqrt{a^2-b}}{2}}$$

Important formulae:

(i) $a^{-P} = \frac{1}{a^P}$ (ii) If $a^y = n$ then $a = (n)^{1/y}$ (iii) If $a^x = b^y$ then $a = (b)^{y/x}$

(iv) $x^n = a,\ x = \sqrt[n]{a}$ (v) $\sqrt[n]{a} = a^{1/n}$ (vi) $\left(\sqrt[n]{a}\right)^m = a^{\frac{m}{n}}$

(vii) $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$

Types of Questions

1. The greatest among $\sqrt{7}-\sqrt{5}, \sqrt{5}-\sqrt{3}$, $\sqrt{9}-\sqrt{7}, \sqrt{11}-\sqrt{9}$

Sol. On rationalising

$$\frac{(\sqrt{7}-\sqrt{5})\times(\sqrt{7}+\sqrt{5})}{\sqrt{7}+\sqrt{5}}, \frac{(\sqrt{5}-\sqrt{3})\times(\sqrt{5}+\sqrt{3})}{\sqrt{5}+\sqrt{3}},$$

$$\frac{(\sqrt{9}-\sqrt{7})\times(\sqrt{9}+\sqrt{7})}{\sqrt{9}+\sqrt{7}}, \frac{(\sqrt{11}-\sqrt{9})\times(\sqrt{11}+\sqrt{9})}{(\sqrt{11}+\sqrt{9})}$$

$$= \frac{2}{\sqrt{7}+\sqrt{5}}, \frac{2}{\sqrt{5}+\sqrt{3}}, \frac{2}{\sqrt{9}+\sqrt{7}}, \frac{2}{\sqrt{11}+\sqrt{9}}$$

smallest one $= \frac{2}{\sqrt{11}+\sqrt{9}} = \sqrt{11}-\sqrt{9}$

Greatest one $= \frac{2}{\sqrt{5}+\sqrt{3}} = \sqrt{5}-\sqrt{3}$

2. Greatest among the following numbers $\sqrt[3]{9}, \sqrt{3}, \sqrt[4]{16}, \sqrt[6]{80}$

Sol. LCM 3, 2, 4, and 6 = 12

$9^{12/3}, 3^{12/2}, 16^{12/4}$, and $80^{12/6}$

$9^4, 3^6, 16^3, 80^2$

6561, 729, 4096, 6400

i.e. largest one $= \sqrt[3]{9}$.

3. By how much does $\sqrt{12}+\sqrt{18}$ exceed $\sqrt{3}+\sqrt{2}$?

Sol. Required value will be the differnce between $\sqrt{12}+\sqrt{18}$ and $(\sqrt{3}+\sqrt{2})$

$$= (\sqrt{12}+\sqrt{18}) - (\sqrt{3}+\sqrt{2})$$

$$= (2\sqrt{3}+3\sqrt{2}) - (\sqrt{3}+\sqrt{2})$$

$$= \sqrt{3}+2\sqrt{2}$$

4. Is $2^x = 3^y = 6^{-z}$ then $\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$ is equal to

Sol. Let $2^x = 3^y = 6^{-z} = k$

i.e. $2 = k^{1/x}$, $3 = k^{1/y}$, $6 = k^{-1/z}$,

$2 \times 3 = 6$

$k^{1/x} \times k^{1/y} = k^{-1/z}$, $k^{1/x+1/y} = k^{-1/z}$

$$\frac{1}{x}+\frac{1}{y} = -\frac{1}{z}, \quad \frac{1}{x}+\frac{1}{y}+\frac{1}{z} = 0$$

5. Find the value of $3+\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{3}+3}+\frac{1}{\sqrt{3}-3}$

Sol. $3+\frac{1\times\sqrt{3}}{\sqrt{3}\times\sqrt{3}}+\frac{1}{(\sqrt{3}+3)}\times\frac{3-\sqrt{3}}{3-\sqrt{3}}+\frac{1}{(\sqrt{3}-3)}\times\frac{3+\sqrt{3}}{3+\sqrt{3}}$

$$= 3+\frac{\sqrt{3}}{3}+\frac{3-\sqrt{3}}{6}-\frac{3+\sqrt{3}}{6} = \frac{18}{6} = 3$$

6. The value of $\sqrt{5+2\sqrt{6}}-\frac{1}{\sqrt{5+2\sqrt{6}}}$

Sol. Expression $\sqrt{5+2\sqrt{6}}-\frac{1}{\sqrt{5+2\sqrt{6}}}$

$$= \frac{\left(\sqrt{5+2\sqrt{6}}\right)^2-1}{\sqrt{5+2\sqrt{6}}} = \frac{5+2\sqrt{6}-1}{\sqrt{5+2\sqrt{6}}}$$

$$= \frac{4+2\sqrt{6}}{\sqrt{3}+\sqrt{2}}\times\left(\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}}\right)$$

$$= 4\sqrt{3}+6\sqrt{2}-4\sqrt{2}-4\sqrt{3} = 2\sqrt{2}$$

7. If a = 64 and b = 289, then the value of $\left(\sqrt{\sqrt{a}+\sqrt{b}}-\sqrt{\sqrt{b}-\sqrt{a}}\right)^{1/2}$

Sol. $\left(\sqrt{\sqrt{64}+\sqrt{289}}-\sqrt{-\sqrt{64}+\sqrt{289}}\right)^{1/2}$

$$\left(\sqrt{8+17}-\sqrt{-8+17}\right)^{1/2} = (5-3)^{1/2} = \sqrt{2}$$

Foundation

Questions

1. If $3^x - 3^{x-1} = 18$, then x^x is equal to
   (a) 3 (b) 8
   (c) 27 (d) 216

2. If $a^{2x+2} = 1$, where a is a positive real number other than 1, then x = ?
   (a) – 2 (b) – 1
   (c) 0 (d) 1

3. $\dfrac{\left[(12)^{-2}\right]^2}{\left[(12)^2\right]^{-2}} = ?$
   (a) 12 (b) 4.8
   (c) $\dfrac{12}{144}$ (d) 1

4. Value of ? in expression
   $7^{8.9} \div (343)^{1.7} \times (49)^{4.8} = 7^?$ is
   (a) 13.4 (b) 12.8
   (c) 11.4 (d) 9.6

5. If $\{(2^4)^{1/2}\}^? = 256$, find the value of '?'.
   (a) 1 (b) 2
   (c) 4 (d) 8

6. $(16)^9 \div (16)^4 \times 16^3 = (16)^?$
   (a) 6.75 (b) 8
   (c) 10 (d) 12

7. $(42 \times 229) \div (9261)^{1/3} = ?$
   (a) 448 (b) 452
   (c) 456 (d) 458

8. Evaluate $(0.00032)^{2/5}$.
   (a) $\dfrac{1}{625}$ (b) $\dfrac{1}{225}$
   (c) $\dfrac{1}{125}$ (d) $\dfrac{1}{25}$

9. If $\dfrac{\sqrt{7}-\sqrt{5}}{\sqrt{7}+\sqrt{5}} = a + b\sqrt{35}$, then the value of (a – b) is:
   (a) 5 (b) 6
   (c) 8 (d) None of these

10. If P = 124, then $\sqrt[3]{P(P^2+3P+3)+1} = ?$
    (a) 5 (b) 7
    (c) 123 (d) 125

11. If $\left(\dfrac{p}{q}\right)^{n-1} = \left(\dfrac{q}{p}\right)^{n-3}$, then the value of n is:
    (a) $\dfrac{1}{2}$ (b) $\dfrac{7}{2}$
    (c) 1 (d) 2

12. Value of ? in $\sqrt[3]{512} \div \sqrt[4]{16} + \sqrt{576} = ?$ is:
    (a) 24 (b) 31
    (c) 28 (d) 18

13. Value of $3 + \dfrac{1}{\sqrt{3}} + \dfrac{1}{3+\sqrt{3}} + \dfrac{1}{\sqrt{3}-3} - 3$ is:
    (a) $3+\sqrt{3}$ (b) 3
    (c) 1 (d) 0

14. $16^{5/4} = ?$
    (a) 64 (b) 31
    (c) 32 (d) 33

15. $\left(\dfrac{32}{243}\right)^{-3/5} = ?$
    (a) $\dfrac{27}{8}$ (b) $\dfrac{27}{7}$
    (c) $\dfrac{27}{6}$ (d) $\dfrac{27}{2}$

16. Find the value of $(243)^{0.16} \times (243)^{0.04}$
    (a) 0.16 (b) $\dfrac{1}{3}$
    (c) 3 (d) 0.04

17. $17^{3.5} \times 17^{7.3} \div 17^{4.2} = 17^?$
    (a) 8.4 (b) 8
    (c) 6.6 (d) 6.4

18. If $289 = 17^{\frac{x}{5}}$, then x = ?
    (a) 16 (b) 8
    (c) 10 (d) $\dfrac{2}{5}$

19. $\left[\left\{\left(-\dfrac{1}{2}\right)^2\right\}^{-2}\right]^{-1} = ?$
    (a) $\dfrac{1}{16}$ (b) 16
    (c) $-\dfrac{1}{16}$ (d) – 16

20. Find the value of $(10)^{200} \div (10)^{196}$.
(a) 10000 (b) 1000
(c) 100 (d) 100000

21. If $\left(\frac{1}{5}\right)^{3a} = 0.008$, find the value of $(0.25)^a$.
(a) 20.5 (b) 22.5
(c) 0.25 (d) 6.25

22. Value of $\sqrt{5+2\sqrt{6}} - \frac{1}{\sqrt{5-2\sqrt{6}}}$ is:
(a) $2\sqrt{2}$ (b) 0
(c) $2\sqrt{3}$ (d) $\sqrt{5}-1$

23. $\frac{1}{1+x^{b-a}+x^{c-a}} + \frac{1}{1+x^{a-b}+x^{c-b}} + \frac{1}{1+x^{b-c}+x^{a-c}} = ?$
(a) x^{a-b-c} (b) 1
(c) 0 (d) 3

24. $[P^{(b-c)}]^{(b+c)} \cdot [P^{(c-a)}]^{(c+a)} \cdot [P^{(a-b)}]^{(a+b)} = ?$
(a) 0 (b) P^{abc}
(c) 1 (d) P^{a+b+c}

25. If $\sqrt{5+\sqrt[3]{x}} = 3$, then the value of x is:
(a) 64 (b) 125
(c) 9 (d) 27

26. $\left[\left(\sqrt[5]{x^{\frac{-3}{5}}}\right)^{-\frac{5}{3}}\right]^5 = ?$
(a) x^5 (b) x^{-5}
(c) x (d) $\frac{1}{x}$

27. $\left(\sqrt{56+\sqrt{56+\sqrt{56+\ldots}}}\right) \div 2^2 = ?$
(a) 0 (b) 1
(c) 2 (d) 8

28. If $a^x = b$, $b^y = c$ and $xyz = 1$, then what is the value of c^z?
(a) a (b) b
(c) ab (d) $\frac{a}{b}$

29. If $16 \times 8^{n+2} = 2^m$, then m is equal to:
(a) n + 8 (b) 2n + 10
(c) 3n + 2 (d) 3n + 10

30. If $x = \frac{\sqrt{3}+1}{\sqrt{3}-1}$ and $y = \frac{\sqrt{3}-1}{\sqrt{3}+1}$, then $x^2 + y^2$ is equal to:
(a) 14 (b) 13
(c) 15 (d) 10

Moderate

1. $\frac{16 \times 2^{n+1} - 4 \times 2^n}{16 \times 2^{n+2} - 2 \times 2^{n+2}} = ?$
(a) 1 (b) 2
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

2. $\frac{2^{2^3} \div (2^2)^3 \times 2^{-2}}{4^{2^3} \div (4^2)^3 \times 4^{-2}} = ?$
(a) 2 (b) $\frac{1}{2}$
(c) 1 (d) – 2

3. $\frac{(0.6)^0 - (0.1)^{-1}}{\left(\frac{3}{2^3}\right)^{-1}\left(\frac{3}{2}\right)^3 + \left(-\frac{1}{3}\right)^{-1}} + \left(\frac{8}{27}\right)^{-\frac{1}{3}} = ?$
(a) 1 (b) 0
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

4. If $\frac{(81)^{4x} \times (27)^x \times 9^7}{(729)^{x+2}} = 3^9$, then find the value of x.
(a) $\frac{6}{13}$ (b) $\frac{5}{13}$
(c) $\frac{8}{13}$ (d) $\frac{7}{13}$

5. If $2^a + 3^b = 17$ and $2^{a+2} - 3^{b+1} = 5$, the values of a and b are :
(a) a = 2, b = 3 (b) a = 4, b = 6
(c) a = 3, b = 2 (d) a = – 3, b = – 2

6. $\left[\frac{3\sqrt{2}}{\sqrt{6}+\sqrt{3}} - \frac{4\sqrt{3}}{\sqrt{6}+\sqrt{2}} + \frac{\sqrt{6}}{\sqrt{3}+\sqrt{2}}\right]$ is simplified to
(a) 0 (b) 1
(c) $\sqrt{3}$ (d) $\sqrt{6}$

7. Find the value of $\frac{2(\sqrt{2}+\sqrt{6})}{3\sqrt{2+3}}$.
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $-\frac{2}{}$ (d) $\frac{4}{}$

8. If $x = \frac{\sqrt{5}-2}{\sqrt{5}+2}$, then $x^4 + x^{-4}$ is
(a) a surd
(b) a rational number but not an integer
(c) an integer
(d) an irrational number but not a surd

9. The simplified value of $\sqrt{\frac{19+8\sqrt{3}}{7-4\sqrt{3}}}$ is :
(a) $11-6\sqrt{3}$ (b) $11+6\sqrt{3}$
(c) $10+5\sqrt{3}$ (d) $10-5\sqrt{3}$

10. The simplified value of $\left(28+10\sqrt{3}\right)^{\frac{1}{2}} - \left(7-4\sqrt{3}\right)^{-\frac{1}{2}}$ is:
(a) 1 (b) 2
(c) 3 (d) 4

11. The simplified value of $\sqrt{6-4\sqrt{3}+\sqrt{16-8\sqrt{3}}}$ is :
(a) $\sqrt{3}$ (b) $\sqrt{3}+1$
(c) $\sqrt{3}-1$ (d) $\sqrt{3}+\sqrt{2}$

12. If $x = \sqrt{\frac{5+2\sqrt{6}}{5-2\sqrt{6}}}$, then the value of $x^2(x-10)^2$ is
(a) 0 (b) 1
(c) – 1 (d) 2

13. The simplified value of $\frac{\sqrt{4-\sqrt{7}}}{\sqrt{8+3\sqrt{7}}-2\sqrt{2}}$ is:
(a) 1 (b) 2
(c) – 1 (d) – 2

14. $\frac{(625)^{6.25} \times (25)^{2.6}}{(625)^{6.75} \times (5)^{1.2}} = ?$
(a) 5 (b) 10
(c) 15 (d) 25

15. The simplified value of $\frac{2+\sqrt{3}}{2-\sqrt{3}}$ is :
(a) $7+4\sqrt{3}$ (b) $7-4\sqrt{3}$
(c) $7+2\sqrt{3}$ (d) $7-2\sqrt{3}$

16. The simplified value of $\frac{(0.3)^{\frac{1}{3}}\left(\frac{1}{27}\right)^{\frac{1}{4}}(9)^{\frac{1}{6}}(0.81)^{\frac{2}{3}}}{(0.9)^{\frac{2}{3}}(3)^{-\frac{1}{2}}(243)^{-\frac{1}{4}}}$ is :
(a) 2.2 (b) 2.7
(c) 2.4 (d) 2.6

17. $\frac{2^{n+4} - 2\times 2^{n}}{2\times 2^{n+3}} + 2^{-3} = ?$
(a) 1 (b) 2
(c) $\frac{1}{2}$ (d) $\frac{1}{3}$

18. If $10^{0.48} = x$, $10^{0.7} = y$ and $x^z = y^2$, then z = ?
(a) $1\frac{11}{12}$ (b) $2\frac{11}{12}$
(c) $3\frac{11}{12}$ (d) $-2\frac{11}{12}$

19. The simplified value of $(27)^{-\frac{2}{3}} + \left[\left(2^{-\frac{2}{3}}\right)^{-\frac{5}{3}}\right]^{-\frac{9}{10}}$ is :
(a) $\frac{11}{18}$ (b) $-\frac{11}{18}$
(c) $\frac{17}{18}$ (d) $-\frac{17}{18}$

20. Find the value of $\sqrt{8+3\sqrt{7}} - \sqrt{7+3\sqrt{5}}$.
(a) $\frac{\sqrt{14}+\sqrt{10}}{2}$ (b) $\frac{\sqrt{14}-\sqrt{10}}{2}$
(c) $\frac{\sqrt{14}+\sqrt{10}}{4}$ (d) $\frac{\sqrt{14}-\sqrt{10}}{4}$

21. Find the value of $\sqrt{38+5\sqrt{3}} + \sqrt{3-\sqrt{5}}$.
(a) $\frac{5\sqrt{6}-\sqrt{10}}{2}$ (b) $\frac{5\sqrt{6}+\sqrt{10}}{2}$
(c) $\frac{5\sqrt{6}-\sqrt{10}}{4}$ (d) $\frac{5\sqrt{6}+\sqrt{10}}{4}$

22. Find the value of $\sqrt{4-\sqrt{7}} + \sqrt{8+3\sqrt{7}}$.
(a) $\sqrt{2}\left(\sqrt{7}+1\right)$ (b) $\sqrt{2}+1$
(c) 1 (d) $\sqrt{2}+2$

23. If $\sqrt{5} = 2.236$ and $\sqrt{10} = 3.162$, then the value of $\frac{15}{\sqrt{10}+\sqrt{20}+\sqrt{40}-\sqrt{5}-\sqrt{80}}$ is:
(a) 4.398 (b) 5.398
(c) 4.938 (d) 5.938

24. $\frac{\left(2^{2n}-3.2^{2n-2}\right)\left(3^{n}-2.3^{n-2}\right)}{3^{n-4}\left(4^{n+3}-2^{2n}\right)} = ?$
(a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) $-\frac{1}{}$ (d) $\frac{1}{6}$

25. If $\frac{\sqrt{2}+\sqrt{3}}{3\sqrt{2}-2\sqrt{3}}=a-b\sqrt{6}$, then the value of $(a+b)^2$ is :

(a) $\frac{13}{36}$ (b) $\frac{49}{36}$

(c) $\frac{7}{6}$ (d) $\frac{16}{39}$

26. The simplified value

$$\frac{7\sqrt{3}}{\sqrt{10}+\sqrt{3}}-\frac{2\sqrt{5}}{\sqrt{6}+\sqrt{5}}-\frac{3\sqrt{2}}{\sqrt{15}+3\sqrt{2}} \text{ is :}$$

(a) 1 (b) 2
(c) 4 (d) 6

27. The simplified value $\frac{\sqrt{5}-2}{\sqrt{5}+2}-\frac{\sqrt{5}+2}{\sqrt{5}-2}$ is :

(a) $8\sqrt{5}$ (b) $-8\sqrt{5}$
(c) $4\sqrt{5}$ (d) $-4\sqrt{5}$

28. The simplified value of

$$\frac{1}{3-\sqrt{8}}-\frac{1}{\sqrt{8}-\sqrt{7}}+\frac{1}{\sqrt{7}-\sqrt{6}}-\frac{1}{\sqrt{6}-\sqrt{5}}+\frac{1}{\sqrt{5}-2} \text{ is :}$$

(a) 3 (b) 4
(c) 5 (d) 6

29. $(9)^{-3}\times\frac{(16)^{\frac{1}{4}}}{(6)^{-2}}\times\left(\frac{1}{27}\right)^{-\frac{4}{3}}=?$

(a) 2 (b) 4
(c) 6 (d) 8

30. $\sqrt{7\sqrt{7\sqrt{7.............\infty}}}=?$

(a) 7 (b) – 7
(c) 4 (d) – 4

Difficult

1. If $x=\frac{\sqrt{3}}{2}$, then $\frac{\sqrt{1+x}}{1+\sqrt{1+x}}+\frac{\sqrt{1-x}}{1-\sqrt{1-x}}$ is equal to

(a) 1 (b) $\frac{2}{\sqrt{3}}$

(c) $2-\sqrt{3}$ (d) 2

2. If $x=\frac{2\sqrt{24}}{\sqrt{3}+\sqrt{2}}$, then the value of $\frac{x+\sqrt{8}}{x-\sqrt{8}}+\frac{x+\sqrt{12}}{x-\sqrt{12}}$ is:

(a) 1 (b) 2
(c) – 1 (d) – 2

3. If $x=\frac{2\sqrt{6}}{\sqrt{3}+\sqrt{2}}$, then the value of $\frac{x+\sqrt{2}}{x-\sqrt{2}}+\frac{x+\sqrt{3}}{x-\sqrt{3}}$ is:

(a) $\sqrt{2}$ (b) $\sqrt{3}$

(c) $\sqrt{6}$ (d) 2

4. If $x=\frac{\sqrt{3}}{2}$, then the value of $\sqrt{1+x}+\sqrt{1-x}$ will be

(a) $\frac{1}{\sqrt{3}}$ (b) $2\sqrt{3}$

(c) $\sqrt{3}$ (d) 2

5. $\sqrt{11-2\sqrt{30}}+\sqrt{7-2\sqrt{10}}-\frac{4}{\sqrt{6}+\sqrt{2}}=?$

(a) 0 (b) 3
(c) – 1 (d) 2

6. $$\sqrt{\frac{(2.4)^6+9(5.76)+6(2.4)^4}{(2.4)^4+6(5.76)+9}}+\frac{\left[(3^{-2})^{-5}\right]^{\frac{1}{5}}+\left[(4^{-3})^{-6}\right]^{\frac{1}{6}}-(3^{-4})^{-\frac{1}{2}}}{\left[(2^{-3})^{-4}\right]^{\frac{1}{4}}}=?$$

(a) 10.2 (b) 9.4
(c) 10.4 (d) 10.8

7. If x is a positive real number and the index be a rational number, then

$$\left(\frac{x^a}{x^{-b}}\right)^{a^2+b^2-ab}\times\left(\frac{x^b}{x^{-c}}\right)^{b^2+c^2-bc}\times\left(\frac{x^c}{x^{-a}}\right)^{c^2+a^2-ca}=?$$

(a) $x^{a^3+b^3+c^3}$ (b) $x^{a^2+b^2+c^2}$

(c) $x^2(a^3+b^3)$ (d) $x^{2(a^3+b^3+c^3)}$

8. If $2^x=4^y=8^z$ and $\frac{1}{2x}+\frac{1}{4y}+\frac{1}{4z}=4$, then x = ?

(a) $\frac{7}{15}$ (b) $\frac{7}{16}$

(c) $\frac{8}{15}$ (d) $\frac{1}{2}$

9. Find the value of $\frac{\sqrt{\sqrt{5}+2}+\sqrt{\sqrt{5}-2}}{\sqrt{\sqrt{5}+1}}-\sqrt{3-2\sqrt{2}}$

(a) $\sqrt{2}$ (b) $\sqrt{2}+1$

(c) 1 (d) $\sqrt{2}+2$

10. $\sqrt{11+4\sqrt{7}}-\sqrt{11-4\sqrt{7}}+\dfrac{\sqrt{8}}{\sqrt{33+10\sqrt{8}}-\sqrt{33-10\sqrt{8}}}$

$=?$

(a) $2\frac{1}{2}$ (b) $3\frac{1}{2}$

(c) $1\frac{1}{2}$ (d) $4\frac{1}{2}$

11. The simplified value of

$\dfrac{26-15\sqrt{3}}{\left[5\sqrt{2}-\sqrt{38+5\sqrt{3}}\right]^2}+\dfrac{\sqrt{10}+\sqrt{18}}{\sqrt{8}+\sqrt{(3-\sqrt{5})}}$ is :

(a) $1\frac{1}{3}$ (b) $2\frac{1}{3}$

(c) $3\frac{1}{3}$ (d) $4\frac{1}{3}$

12. The simplified value of

$(28-10\sqrt{3})^{\frac{1}{2}}-(7+4\sqrt{3})^{-\frac{1}{2}}+\dfrac{\sqrt{7}}{\sqrt{16+6\sqrt{7}}-\sqrt{16-6\sqrt{7}}}$

is:

(a) $1\frac{1}{2}$ (b) $2\frac{1}{2}$

(c) $3\frac{1}{2}$ (d) $4\frac{1}{2}$

13. $\dfrac{4\sqrt{3}}{2-\sqrt{2}}-\dfrac{30}{4\sqrt{3}-\sqrt{18}}-\dfrac{\sqrt{18}}{3-2\sqrt{3}}=?$

(a) $2\sqrt{6}$ (b) $\sqrt{6}$

(c) $3\sqrt{6}$ (d) $4\sqrt{6}$

14. The simplified value of

$\sqrt{\dfrac{(12.12)^2-(8.12)^2}{(0.25)^2+(0.25)\times(19.99)}}+\dfrac{\left[\left(8^{-\frac{3}{4}}\right)^{\frac{5}{2}}\right]^{\frac{8}{15}}\times 16^{\frac{3}{4}}}{\sqrt[3]{\left[\left\{(128^{-5})^{\frac{3}{7}}\right\}^{-\frac{1}{5}}\right]}}$ is :

(a) $3\frac{1}{2}$ (b) $2\frac{1}{2}$

(c) $4\frac{1}{2}$ (d) $5\frac{1}{2}$

15. $\dfrac{\sqrt[6]{2}\left[(625)^{\frac{3}{5}}\times(1024)^{-\frac{6}{5}}\div(25)^{\frac{3}{5}}\right]^{\frac{1}{2}}}{\left(\sqrt[3]{128}\right)^{-\frac{5}{2}}\times(125)^{\frac{1}{5}}}+\dfrac{(10^3)^2\div\left(10^{3^2}\right)}{(10^2)^3\div 10^{2^3}}$

(a) 1.2 (b) 1.4

(c) 1.6 (d) 1.1

16. $(18-8\sqrt{2})^{\frac{1}{2}}+(6-4\sqrt{2})^{-\frac{1}{2}}+\dfrac{\sqrt{3}}{\sqrt{19+8\sqrt{3}}-\sqrt{19-8\sqrt{3}}}=$

(a) $\dfrac{11-\sqrt{2}}{2}$ (b) $\dfrac{11+\sqrt{2}}{2}$

(c) $\dfrac{11-2\sqrt{2}}{2}$ (d) $\dfrac{11+2\sqrt{2}}{2}$

17. $\sqrt{7+4\sqrt{3}}-\sqrt{28+10\sqrt{3}}+\dfrac{\sqrt{11}}{\sqrt{20+6\sqrt{11}}+\sqrt{20-6\sqrt{11}}}=$

(a) $2\frac{1}{2}$ (b) $1\frac{1}{2}$

(c) $-1\frac{1}{2}$ (d) $-2\frac{1}{2}$

18. $\dfrac{1}{\sqrt{11-2\sqrt{30}}}-\dfrac{3}{\sqrt{7-2\sqrt{10}}}-\dfrac{4}{\sqrt{8+4\sqrt{3}}}=?$

(a) 0 (b) 1

(c) – 1 (d) $-2\sqrt{3}$

19. If $x=\dfrac{\sqrt{3}}{2}$, then $\dfrac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}-\sqrt{1-x}}=?$

(a) $-\sqrt{3}$ (b) $\sqrt{3}$

(c) $-\sqrt{2}$ (d) $\sqrt{2}$

20. If $x=\dfrac{\sqrt{a+2b}+\sqrt{a-2b}}{\sqrt{a+2b}-\sqrt{a-2b}}$, then the value of bx^2-ax+b is :

(a) 0 (b) – 1

(c) 1 (d) – 2

Previous Year (Memory Based)

1. If $\left(\frac{3}{5}\right)^3\left(\frac{3}{5}\right)^{-6}=\left(\frac{3}{5}\right)^{2x-1}$, then x is equal to

(a) – 2 (b) 2
(c) – 1 (d) 1

2. If $\frac{\sqrt{7}-2}{\sqrt{7}+2}=a\sqrt{7}+b$, then the value of a is

(a) $\frac{11}{3}$ (b) $-\frac{4}{3}$
(c) $\frac{4}{3}$ (d) $-\frac{4\sqrt{7}}{3}$

3. If $x=\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$ and $y=\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$ then (x + y) equals:

(a) 8 (b) 16
(c) $2\sqrt{15}$ (d) $2(\sqrt{5}+\sqrt{3})$

4. If $\sqrt{2}=1.414$, the square root of $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ is nearest to

(a) 0.172 (b) 0.414
(c) 0.586 (d) 1.414

5. For what value(s) of a is $x+\frac{1}{4}\sqrt{x}+a^2$ is a perfect square?

(a) $\pm\frac{1}{18}$ (b) $\frac{1}{8}$
(c) $-\frac{1}{5}$ (d) $\frac{1}{4}$

6. Which of the following numbers is the least ?

$(0.5)^2$, $\sqrt{0.49}$, $\sqrt[3]{0.008}$, 0.23

(a) $(0.5)^2$ (b) $\sqrt{0.49}$
(c) $\sqrt[3]{0.008}$ (d) 0.23

7. Arrange the following in descending order :

$\sqrt[3]{4}, \sqrt{2}, \sqrt[6]{3}, \sqrt[4]{5}$

(a) $\sqrt[3]{4}>\sqrt[4]{5}>\sqrt{2}>\sqrt[6]{3}$
(b) $\sqrt[4]{5}>\sqrt[3]{4}>\sqrt[6]{3}>\sqrt{2}$
(c) $\sqrt{2}>\sqrt[6]{3}>\sqrt[3]{4}>\sqrt[4]{5}$
(d) $\sqrt{2}>\sqrt[3]{4}>\sqrt[6]{3}>\sqrt[4]{5}$

8. $\left[\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}-\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}\right]$ simplifies to

(a) $2\sqrt{6}$ (b) $4\sqrt{6}$
(c) $2\sqrt{3}$ (d) $3\sqrt{2}$

9. By how much does $(\sqrt{12}+\sqrt{18})$ exceed $(2\sqrt{3}+2\sqrt{2})$

(a) 2 (b) $\sqrt{3}$
(c) $\sqrt{2}$ (d) 3

10. The greatest number among $\sqrt{5}, \sqrt[3]{4}, \sqrt[5]{2}, \sqrt[7]{3}$ is:

(a) $\sqrt[3]{4}$ (b) $\sqrt[7]{3}$
(c) $\sqrt{5}$ (d) $\sqrt[5]{2}$

11. If $\sqrt{7\sqrt{7\sqrt{7\sqrt{7.....}}}}=(343)^{y-1}$, then y is equal to

(a) $\frac{2}{3}$ (b) 1
(c) $\frac{4}{3}$ (d) $\frac{3}{4}$

12. $\frac{1}{\sqrt[3]{4}+\sqrt[3]{2}+1}$ is equal to

(a) $\sqrt[3]{2}+1$ (b) $\sqrt[3]{2}-1$
(c) $-\sqrt[3]{2}-1$ (d) $1-\sqrt[3]{2}$

13. $\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+.....+\frac{1}{\sqrt{99}+\sqrt{100}}$ is equal to

(a) $10-\sqrt{99}$ (b) $\sqrt{2}-10$
(c) 7 (d) 9

14. If $\sqrt{6}\times\sqrt{15}=x\sqrt{10}$, then the value of x is

(a) 3 (b) ± 3
(c) $\sqrt{3}$ (d) $\sqrt{6}$

15. The largest among the numbers

$\sqrt{7}-\sqrt{5}, \sqrt{5}-\sqrt{3}, \sqrt{9}-\sqrt{7}, \sqrt{11}-\sqrt{9}$ is :

(a) $\sqrt{7}-\sqrt{5}$ (b) $\sqrt{5}-\sqrt{3}$
(c) $\sqrt{9}-\sqrt{7}$ (d) $\sqrt{11}-\sqrt{9}$

16. The largest among the numbers $(0.1)^2$, $\sqrt{0.0121}$, 0.12 and $\sqrt{0.0004}$ is :

(a) $(0.1)^2$ (b) $\sqrt{0.0121}$
(c) 0.12 (d) $\sqrt{0.0004}$

17. $\sqrt{6+\sqrt{6+\sqrt{6+.....}}}$ is equal to
(a) 2 (b) 5
(c) 4 (d) 3

18. The value of $\frac{(81)^{3.6}\times(9)^{2.7}}{(81)^{4.2}\times(3)}$ is :
(a) 3 (b) 6
(c) 9 (d) 8.2

19. If $a=\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$, $b=\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$, then the value of $\frac{a^2}{b}+\frac{b^2}{a}$ is :
(a) 900 (b) 970
(c) 1030 (d) 930

20. If $2\sqrt{x}=\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}-\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$, then the value of x is :
(a) 6 (b) 30
(c) $\sqrt{15}$ (d) 15

21. If $\sqrt{3}=1.732$, the value of
$$\frac{3+\sqrt{6}}{5\sqrt{3}-2\sqrt{12}-\sqrt{32}+\sqrt{50}}$$ is :
(a) 4.899 (b) 2.551
(c) 1.414 (d) 1.732

22. $\sqrt{10+2\sqrt{6}+2\sqrt{10}+2\sqrt{15}}$ is equal to
(a) $(\sqrt{2}+\sqrt{3}-\sqrt{5})$ (b) $(\sqrt{3}+\sqrt{5}-\sqrt{2})$
(c) $(\sqrt{2}+\sqrt{5}-\sqrt{3})$ (d) $(\sqrt{2}+\sqrt{3}+\sqrt{5})$

23. The value of $\sqrt{\frac{(\sqrt{12}-\sqrt{8})(\sqrt{3}+\sqrt{2})}{5+\sqrt{24}}}$ is :
(a) $\sqrt{6}-\sqrt{2}$ (b) $\sqrt{6}+\sqrt{2}$
(c) $\sqrt{6}-2$ (d) $2-\sqrt{6}$

24. If $a=\frac{2+\sqrt{3}}{2-\sqrt{3}}$ and $b=\frac{2-\sqrt{3}}{2+\sqrt{3}}$, then the value of (a^2+b^2+ab)
(a) 169 (b) 195
(c) 168 (d) 196

25. $\frac{\sqrt{7}}{\sqrt{16+6\sqrt{7}}-\sqrt{16-6\sqrt{7}}}$ is equal to
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{1}{4}$ (d) $\frac{1}{5}$

26. By how much does $\sqrt{12}+\sqrt{18}$ exceed $\sqrt{3}+\sqrt{2}$?
(a) $2(\sqrt{3}-\sqrt{2})$ (b) $2(\sqrt{3}+\sqrt{2})$
(c) $\sqrt{3}+2\sqrt{2}$ (d) $\sqrt{2}-4\sqrt{3}$

27. When simplified $(256)^{-(4^{-3/2})}$ is
(a) 8 (b) $\frac{1}{8}$
(c) 2 (d) $\frac{1}{2}$

28. The largest among the numbers 0.9, $\sqrt{0.9}$ $0.\bar{9}$ is
(a) 0.9 (b) $(0.9)^2$
(c) $\sqrt{0.9}$ (d) $0.\bar{9}$

29. $\sqrt{\sqrt[3]{0.004096}}$ is equal to
(a) 4 (b) 0.4
(c) 0.04 (d) 0.004

30. The ascending order of $(2.89)^{0.5}$, $2-(0.5)^2$, $\sqrt{3}$ and $\sqrt[3]{0.008}$ is
(a) $2-(0.5)^2, \sqrt{3}, \sqrt[3]{0.008}, (2.89)^{05}$
(b) $\sqrt[3]{0.008}, (2.89)^{0.5}, \sqrt{3}, 2-(0.5)^2$
(c) $\sqrt[3]{0.008}, \sqrt{3}, (2.89)^{05}, 2-(0.5)^2$
(d) $\sqrt{3}, \sqrt[3]{0.008}, 2-(0.5)^2, (2.89)^{05}$

Foundation

Solutions

1. **(c);** $3^{x-1}(3-1)=18$
$3^{x-1}=9=3^2$
$x-1=2 \Rightarrow x=3$ so, $x^x=27$

2. **(b);** $a^{2x+2}=a^{\circ}$
$2x=-2 \Rightarrow x=-1$

3. **(d);** $\frac{12^{-4}}{12^{-4}}=1$

4. **(a);** $7^{?} = \dfrac{7^{8.9} \times (7^2)^{4.8}}{(7^3)^{1.7}} = \dfrac{7^{18.5}}{7^{5.1}} = 7^{13.4}$

$? = 13.4$

5. **(c);** $2^{4 \times \frac{1}{2} \times ?} = 2^8$

$? = 4$

6. **(b);** $(16)^{?} = \dfrac{(16)^9 \times (16)^3}{(16)^4} = 16^{9+3-4} = 16^8$

$? = 8$

7. **(d);** $\dfrac{(42 \times 229)}{(9261)^{\frac{1}{3}}} = \dfrac{42 \times 229}{21} = 458$

8. **(d);** $\left(\dfrac{32}{100000}\right)^{\frac{2}{5}} = \left\{\left(\dfrac{2}{10}\right)^5\right\}^{\frac{2}{5}} = \left(\dfrac{1}{5}\right)^2 = \dfrac{1}{25}$

9. **(d);** $\dfrac{\sqrt{7}-\sqrt{5}}{\sqrt{7}+\sqrt{5}} = \dfrac{(\sqrt{7}-\sqrt{5})(\sqrt{7}-\sqrt{5})}{2} = 6 - \sqrt{35}$

on comparision, $a = 6$ and $b = -1$

$a - b = 6 - (-1) = 7$

10. **(d);** $\sqrt[3]{P(P^2+3P+3)+1} = \sqrt[3]{P^3+3P^2+3P+1}$

$= \sqrt[3]{(P+1)^3} = P+1$

$P = 124, \quad = P+1 = 125$

11. **(d);** $\left(\dfrac{p}{q}\right)^{n-1} = \left(\dfrac{p}{q}\right)^{3-n}$

$n - 1 = 3 - n, \quad 2n = 4 \Rightarrow n = 2$

12. **(c);** $? = 8 \div 2 + 24 = 28$

13. **(d);** $3 + \dfrac{1}{\sqrt{3}} + \dfrac{(3-\sqrt{3})}{(3+\sqrt{3})(3-\sqrt{3})} + \dfrac{(\sqrt{3}+3)}{(\sqrt{3}-3)(\sqrt{3}+3)} - 3$

$= 3 + \dfrac{1}{\sqrt{3}} + \dfrac{3-\sqrt{3}-\sqrt{3}-3}{6} - 3 \; = 3 + \dfrac{1}{\sqrt{3}} - \dfrac{1}{\sqrt{3}} - 3 = 0$

14. **(c);** $(16)^{\frac{5}{4}} = (2^4)^{\frac{5}{4}} = 32$

15. **(a);** $\left(\dfrac{243}{32}\right)^{\frac{3}{5}} = \left(\dfrac{3^5}{2^5}\right)^{\frac{3}{5}} = \dfrac{27}{8}$

16. **(c);** $(243)^{0.2} = (3^5)^{0.2} = 3$

17. **(c);** $17^{3.5+7.3-4.2} = 17^x$

$x = 6.6$

18. **(c);** $17^2 = 17^{\frac{x}{5}} \Rightarrow \dfrac{x}{5} = 2 \Rightarrow x = 10$

19. **(a);** $\left[\left\{\dfrac{1}{4}\right\}^{-2}\right]^{-1} = \left\{\dfrac{1}{\frac{1}{16}}\right\}^{-1} = 16^{-1} = \dfrac{1}{16}$

20. **(a);** $10^{200-196} = 10000$

21. **(c);** $\left(\dfrac{1}{5}\right)^{3a} = 0.008 = \dfrac{8}{1000} = \left(\dfrac{1}{5}\right)^3$

$3a = 3 \Rightarrow a = 1$

$(0.25)^a = 0.25$

22. **(b);** $\sqrt{(\sqrt{3})^2+(\sqrt{2})^2+2\sqrt{3}\sqrt{2}} - \dfrac{1}{\sqrt{(\sqrt{3})^2+(\sqrt{2})^2-2\sqrt{3}\sqrt{2}}}$

$= \sqrt{3}+\sqrt{2} - \dfrac{1}{\sqrt{3}-\sqrt{2}} = \sqrt{3}+\sqrt{2}-\sqrt{3}-\sqrt{2} = 0$

23. **(b);** $\dfrac{1}{1+\frac{x^b}{x^a}+\frac{x^c}{x^a}} + \dfrac{1}{1+\frac{x^a}{x^b}+\frac{x^c}{x^b}} + \dfrac{1}{1+\frac{x^b}{x^c}+\frac{x^a}{x^c}}$

$= \dfrac{x^a}{x^a+x^b+x^c} + \dfrac{x^b}{x^a+x^b+x^c} + \dfrac{x^c}{x^a+x^b+x^c} = 1$

24. **(c);** $P^{b^2-c^2} \cdot P^{c^2-a^2} \cdot P^{a^2-b^2} = P^{b^2-c^2+c^2-a^2+a^2-b^2} = 1$

25. **(a);** $\sqrt{5+\sqrt[3]{x}} = 3$

$\sqrt[3]{x} = 9 - 5 \Rightarrow x = 4^3 = 64$

26. **(c);** $\left[\left\{x^{-\frac{3}{5} \times \frac{1}{5}}\right\}^{-\frac{5}{3}}\right]^5 = x^{\frac{-3}{5} \times \frac{1}{5} \times \frac{-5}{3} \times 5} = x$

27. **(c);** $56 = 8 \times 7$

so, $\sqrt{56+\sqrt{56+\sqrt{56+.....}}} = 8$

$= 8 \div 2^2 = 2$

28. **(a);** $b = a^x$

$b^y = a^{xy}$

$b^{yz} = a^{xyz}$

$c^z = a$

29. **(d);** $2^4 \times 2^{3(n+2)} = 2^m$

$2^m = 2^{(3n+10)}$

$3n + 10 = m$

30. **(a);** $x^2 + y^2 = \left(\dfrac{\sqrt{3}+1}{\sqrt{3}-1}\right)^2 + \left(\dfrac{\sqrt{3}-1}{\sqrt{3}+1}\right)^2$

$= \dfrac{4+2\sqrt{3}}{4-2\sqrt{3}} + \dfrac{4-2\sqrt{3}}{4+2\sqrt{3}}$

$= \dfrac{(4+2\sqrt{3})^2 + (4-2\sqrt{3})^2}{(4-2\sqrt{3})(4+2\sqrt{3})}$

$= \dfrac{16+12+16\sqrt{3}+16+12-16\sqrt{3}}{16-12} = \dfrac{56}{4} = 14$

Moderate

1. (c); Expression = $\dfrac{16\times2^{n+1}-4\times2^{n}}{16\times2^{n+2}-2\times2^{n+2}}$

$$=\frac{2^4\times2^{n+1}-2^2\times2^{n}}{2^4\times2^{n+2}-2\times2^{n+2}}=\frac{2^{n+5}-2^{n+2}}{2^{n+6}-2^{n+3}}$$

$$=\frac{2^{n+5}-2^{n+2}}{2\times2^{n+5}-2\times2^{n+2}}=\frac{2^{n+5}-2^{n+2}}{2(2^{n+5}-2^{n+2})}=\frac{1}{2}$$

2. (c); Expression = $\dfrac{2^{2^3}\div(2^2)^3\times2^{-2}}{4^{2^3}\div(4^2)^3\times4^{-2}}$

$$=\frac{2^{2\times2\times2}\div2^{2\times3}\times2^{-2}}{4^{2\times2\times2}\div4^{2\times3}\times4^{-2}}$$

$$=\frac{2^8\div2^6\times2^{-2}}{4^8\div4^6\times4^{-2}}=\frac{2^8\times\frac{1}{2^6}\times\frac{1}{2^2}}{4^8\times\frac{1}{4^6}\times\frac{1}{4^2}}=1$$

3. (b); Expression

$$=\frac{(0.6)^0-(0.1)^{-1}}{\left(\frac{3}{2^3}\right)^{-1}\left(\frac{3}{2}\right)^3+\left(-\frac{1}{3}\right)^{-1}}+\left(\frac{8}{27}\right)^{-\frac{1}{3}}$$

$$=\frac{1-\left(\frac{1}{10}\right)^{-1}}{\left(\frac{3}{8}\right)^{-1}\left(\frac{3}{2}\right)^3+\left(-\frac{1}{3}\right)^{-1}}+\left(\frac{8}{27}\right)^{-\frac{1}{3}}$$

$$=\frac{1-10}{\frac{8}{3}\times\frac{27}{8}+(-3)}+\left(\frac{27}{8}\right)^{\frac{1}{3}}$$

$$=\frac{-9}{9-3}+\left[\left(\frac{3}{2}\right)^3\right]^{\frac{1}{3}}=-\frac{9}{6}+\frac{3}{2}=-\frac{3}{2}+\frac{3}{2}=0$$

4. (d); $\dfrac{(81)^{4x}\times(27)^x\times9^7}{(729)^{x+2}}=3^9$

$$\Rightarrow\frac{(3^4)^{4x}\times(3^3)^x\times(3^2)^7}{(3^6)^{x+2}}=3^9$$

$$\Rightarrow\frac{3^{16x}\times3^{3x}\times3^{14}}{3^{6x+12}}=3^9$$

$$\Rightarrow\frac{3^{16x+3x+14}}{3^{6x+12}}=3^9\quad\Rightarrow\quad3^{19x+14-6x-12}=3^9$$

$$3^{13x+2}=3^9\quad\Rightarrow\quad13x+2=9$$

$$13x=9-2=7\quad\Rightarrow\quad x=\frac{7}{13}$$

5. (c); The equations are
$2^a + 3^b = 17$ and $2^a \times 2^2 - 3^b \times 3 = 5$
Let $x = 2^a$ and $y = 3^b$
then, $x + y = 17$ ... (i)
and $4x - 3y = 5$ ... (ii)
By equation (i) × 3 + (ii),

$$\begin{array}{rcrcr} 3x & + & 3y & = & 51\\ 4x & - & 3y & = & 5\\ \hline 7x & & & = & 56 \end{array}$$

From equation (i),
$y = 17 - x = 17 - 8 = 9$
$\therefore$ $x = 8$ and $y = 9$
$\therefore$ $x = 2^a = 8 \Rightarrow 2^a = 2^3 \Rightarrow a = 3$
and $y = 3^b = 9 \Rightarrow 3^b = 3^2 \Rightarrow b = 2$
$\therefore$ $a = 3$ and $b = 2$

6. (a); Expression = $\dfrac{3\sqrt{2}}{\sqrt{6}+\sqrt{3}}-\dfrac{4\sqrt{3}}{\sqrt{6}+\sqrt{2}}+\dfrac{\sqrt{6}}{\sqrt{3}+\sqrt{2}}$

$$=\frac{3\sqrt{2}}{\sqrt{6}+\sqrt{3}}\times\frac{\sqrt{6}-\sqrt{3}}{\sqrt{6}-\sqrt{3}}-\frac{4\sqrt{3}}{\sqrt{6}+\sqrt{2}}\times\frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}-\sqrt{2}}$$

$$+\frac{\sqrt{6}}{\sqrt{3}+\sqrt{2}}\times\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}}$$

[Rationalising the respective denominators]

$$=\frac{3\sqrt{2}(\sqrt{6}-\sqrt{3})}{6-3}-\frac{4\sqrt{3}(\sqrt{6}-\sqrt{2})}{6-2}+\frac{\sqrt{6}(\sqrt{3}-\sqrt{2})}{3-2}$$

$$=\sqrt{2}(\sqrt{6}-\sqrt{3})-\sqrt{3}(\sqrt{6}-\sqrt{2})+\sqrt{6}(\sqrt{3}-\sqrt{2})$$

$$=\sqrt{12}-\sqrt{6}-\sqrt{18}+\sqrt{6}+\sqrt{18}-\sqrt{12}=0$$

7. (d); Let $x=\dfrac{2(\sqrt{2}+\sqrt{6})}{3\sqrt{2+\sqrt{3}}}$

Squaring both sides, $x^2=\left[\dfrac{2(\sqrt{2}+\sqrt{6})}{3\sqrt{2+\sqrt{3}}}\right]^2$

$$x^2=\frac{4(\sqrt{2}+\sqrt{6})^2}{\left(3\sqrt{2+\sqrt{3}}\right)^2}\quad\Rightarrow\quad x^2=\frac{4(\sqrt{2}+\sqrt{6})^2}{9(2+\sqrt{3})}$$

$$x^2=\frac{4(2+6+2\times\sqrt{2}\times\sqrt{6})}{9(2+\sqrt{3})}$$

$$x^2 = \frac{4(8+2\sqrt{12})}{9(2+\sqrt{3})} \Rightarrow x^2 = \frac{4(8+2\sqrt{2^2\times 3})}{9(2+\sqrt{3})}$$

$$x^2 = \frac{4(8+4\sqrt{3})}{9(2+\sqrt{3})} \Rightarrow x^2 = \frac{16(2+\sqrt{3})}{9(2+\sqrt{3})} = \frac{16}{9}$$

$$x = \frac{4}{3}$$

8. **(c);** $x = \dfrac{\sqrt{5}-2}{\sqrt{5}+2}$

$$= \frac{(\sqrt{5}-2)^2}{(\sqrt{5}+2)(\sqrt{5}-2)} = \frac{5+4-4\sqrt{5}}{5-4} = 9-4\sqrt{5}$$

$$\therefore \frac{1}{x} = 9+4\sqrt{5}, \quad x^4 + x^{-4} = x^4 + \frac{1}{x^4}$$

$$= \left(x^2 + \frac{1}{x^2}\right)^2 - 2 = \left[\left(x+\frac{1}{x}\right)^2 - 2\right]^2 - 2$$

$$= \left[(9+4\sqrt{5}+9-4\sqrt{5})^2 - 2\right]^2 - 2$$

$$= [(18)^2 - 2]^2 - 2 = (322)^2 - 2 = 103682 \text{ (integer)}$$

9. **(b);** Expression $= \sqrt{\dfrac{19+8\sqrt{3}}{7-4\sqrt{3}}}$

$$= \sqrt{\frac{19+2\times 4\times\sqrt{3}}{7-2\times 2\times\sqrt{3}}} = \sqrt{\frac{16+3+2\times 4\times\sqrt{3}}{4+3-2\times 2\times\sqrt{3}}}$$

$$= \sqrt{\frac{(4)^2+(\sqrt{3})^2+2\times 4\times\sqrt{3}}{(2)^2+(\sqrt{3})^2-2\times 2\times\sqrt{3}}} = \sqrt{\frac{(4+\sqrt{3})^2}{(2-\sqrt{3})^2}}$$

$$\left[\because (a\pm b)^2 = a^2+b^2\pm 2ab\right]$$

$$= \frac{4+\sqrt{3}}{2-\sqrt{3}}$$

Rationalising the denominator.

$$= \frac{(4+\sqrt{3})}{(2-\sqrt{3})}\times\frac{(2+\sqrt{3})}{(2+\sqrt{3})} = \frac{8+2\sqrt{3}+4\sqrt{3}+3}{4-3}$$

$$= 11+6\sqrt{3}$$

10. **(c);** First term $= (28+10\sqrt{3})^{\frac{1}{2}}$

$$= (28+2\times 5\times\sqrt{3})^{\frac{1}{2}}$$

$$= (25+3+2\times 5\times\sqrt{3})^{\frac{1}{2}}$$

$$= \left[(5)^2+(\sqrt{3})^2+2\times 5\times\sqrt{3}\right]^{\frac{1}{2}}$$

$$= \left[(5+\sqrt{3})^2\right]^{\frac{1}{2}} = 5+\sqrt{3}$$

Second term $= (7-4\sqrt{3})^{-\frac{1}{2}} = (7-2\times 2\times\sqrt{3})^{-\frac{1}{2}}$

$$= (4+3-2\times 2\times\sqrt{3})^{-\frac{1}{2}}$$

$$= \left[(2)^2+(\sqrt{3})^2-2\times 2\times\sqrt{3}\right]^{-\frac{1}{2}}$$

$$= \left[(2-\sqrt{3})^2\right]^{-\frac{1}{2}} = (2-\sqrt{3})^{2\times\frac{-1}{2}}$$

$$= (2-\sqrt{3})^{-1} = \frac{1}{(2-\sqrt{3})}$$

$$= \frac{1\times(2+\sqrt{3})}{(2-\sqrt{3})(2+\sqrt{3})} = \frac{(2+\sqrt{3})}{4-3} = 2+\sqrt{3}$$

$\therefore$ Expression

$$= (5+\sqrt{3})-(2+\sqrt{3}) = 5+\sqrt{3}-2-\sqrt{3} = 3$$

11. **(c);** Expression $= \sqrt{6-4\sqrt{3}+\sqrt{16-8\sqrt{3}}}$

$$= \sqrt{6-4\sqrt{3}+\sqrt{16-2\times 2\sqrt{3}\times 2}}$$

$$= \sqrt{6-4\sqrt{3}+\sqrt{12+4-2\times 2\sqrt{3}\times 2}}$$

$$= \sqrt{6-4\sqrt{3}+\sqrt{(2\sqrt{3})^2+(2)^2-2\times 2\sqrt{3}\times 2}}$$

$$= \sqrt{6-4\sqrt{3}+\sqrt{(2\sqrt{3}-2)^2}}$$

$$= \sqrt{6-4\sqrt{3}+2\sqrt{3}-2}$$

$$= \sqrt{4-2\sqrt{3}} = \sqrt{3+1-2\times\sqrt{3}\times 1}$$

$$= \sqrt{(\sqrt{3})^2+(1)^2-2\times\sqrt{3}\times 1}$$

$$= \sqrt{(\sqrt{3}-1)^2} = \sqrt{3}-1$$

12. **(b);** $x = \sqrt{\dfrac{5+2\sqrt{6}}{5-2\sqrt{6}}}$ Rationalising,

$$x = \sqrt{\frac{5+2\sqrt{6}}{5-2\sqrt{6}}\times\frac{5+2\sqrt{6}}{5+2\sqrt{6}}}$$

$$= \sqrt{\frac{(5+2\sqrt{6})^2}{25-24}} = 5+2\sqrt{6}$$

$\therefore\ x^2(x-10)^2$

$= \left(5+2\sqrt{6}\right)^2\left(5+2\sqrt{6}-10\right)^2$

$= \left(5+2\sqrt{6}\right)^2\left(2\sqrt{6}-5\right)^2$

$= \left(25+24+20\sqrt{6}\right)\left(24+25-20\sqrt{6}\right)$

$= \left(49+20\sqrt{6}\right)\left(49-20\sqrt{6}\right)$

$= (49)^2-\left(20\sqrt{6}\right)^2 = 2401-2400 = 1$

13. (a); $\dfrac{\sqrt{4-\sqrt{7}}}{\sqrt{8+3\sqrt{7}}-2\sqrt{2}} = \dfrac{\sqrt{8-2\sqrt{7}}}{\sqrt{16+6\sqrt{7}}-4}$

(Multiplying numerator and denominator by $\sqrt{2}$)

$= \dfrac{\sqrt{(1)^2+\left(\sqrt{7}\right)^2-2\times1\times\sqrt{7}}}{\sqrt{(3)^2+\left(\sqrt{7}\right)^2+2.3.\sqrt{7}}-4}$

$\left\{\therefore 8 = 1+7 = 1^2+\left(\sqrt{7}\right)^2 \text{ and } 16 = 9+7 = 3^2+\left(\sqrt{7}\right)^2\right\}$

$= \dfrac{\sqrt{\left(\sqrt{7}-1\right)^2}}{\sqrt{\left(\sqrt{7}+3\right)^2}-4}$ $\{\because a^2+b^2-2ab=(a-b)^2\}$

$= \dfrac{\sqrt{7}-1}{\sqrt{7}+3-4} = \dfrac{\sqrt{7}-1}{\sqrt{7}-1} = 1$

14. (d); Expression $= \dfrac{(625)^{6.25}\times(25)^{2.6}}{(625)^{6.75}\times(5)^{1.2}}$

$= \dfrac{(5^4)^{6.25}\times(5^2)^{2.6}}{(5^4)^{6.75}\times(5)^{1.2}} = \dfrac{(5)^{4\times6.25}\times(5)^{2\times2.6}}{(5)^{4\times6.75}\times(5)^{1.2}}$

$= \dfrac{(5)^{25}\times(5)^{5.2}}{(5)^{27}\times(5)^{1.2}} = \dfrac{(5)^{25+5.2}}{(5)^{27+1.2}} = \dfrac{(5)^{30.2}}{(5)^{28.2}}$

$= (5)^{30.2-28.2} = (5)^2 = 25$

15. (a); Expression $= \dfrac{2+\sqrt{3}}{2-\sqrt{3}}$

$= \dfrac{\left(2+\sqrt{3}\right)}{\left(2-\sqrt{3}\right)}\times\dfrac{\left(2+\sqrt{3}\right)}{\left(2+\sqrt{3}\right)}$

$\left[\text{Multiplying N}^r \text{ and D}^r \text{ by } \left(2+\sqrt{3}\right)\right]$

$= \dfrac{\left(2+\sqrt{3}\right)^2}{(2)^2-\left(\sqrt{3}\right)^2}$ $[\because (a+b)(a-b)=a^2-b^2]$

$= \dfrac{4+3+2\times2\sqrt{3}}{4-3} = 7+4\sqrt{3}$

16. (b); Expression $= \dfrac{(0.3)^{\frac{1}{3}}\left(\frac{1}{27}\right)^{\frac{1}{4}}(9)^{\frac{1}{6}}(0.81)^{\frac{2}{3}}}{(0.9)^{\frac{2}{3}}(3)^{-\frac{1}{2}}(243)^{-\frac{1}{4}}}$

$= \dfrac{(0.3)^{\frac{1}{3}}(9)^{\frac{1}{6}}(0.81)^{\frac{2}{3}}\times(3)^{\frac{1}{2}}(243)^{\frac{1}{4}}}{(0.9)^{\frac{2}{3}}\times(27)^{\frac{1}{4}}}$

$= \dfrac{(0.3)^{\frac{1}{3}}(3^2)^{\frac{1}{6}}(0.9)^{2\times\frac{2}{3}}\times(3)^{\frac{1}{2}}(3^5)^{\frac{1}{4}}}{(0.9)^{\frac{2}{3}}\times(3^3)^{\frac{1}{4}}}$

$= \dfrac{(0.3)^{\frac{1}{3}}3^{\frac{1}{3}}\times(0.9)^{\frac{4}{3}}\times3^{\frac{1}{2}}\times3^{\frac{5}{4}}}{(0.9)^{\frac{2}{3}}\times3^{\frac{3}{4}}}$

$= (0.3)^{\frac{1}{3}}\times(0.9)^{\frac{4}{3}-\frac{2}{3}}\times3^{\frac{1}{3}+\frac{1}{2}+\frac{5}{4}-\frac{3}{4}}$

$= (0.3)^{\frac{1}{3}}\times(0.9)^{\frac{2}{3}}\times3^{\frac{4+6+15-9}{12}}$

$= (0.3)^{\frac{1}{3}}\times(0.3\times3)^{\frac{2}{3}}\times3^{\frac{16}{12}}$

$= (0.3)^{\frac{1}{3}}\times(0.3)^{\frac{2}{3}}\times(3)^{\frac{2}{3}}\times(3)^{\frac{4}{3}}$

$= (0.3)^{\frac{1}{3}+\frac{2}{3}}\times3^{\frac{2}{3}+\frac{4}{3}} = (0.3)^{\frac{1+2}{3}}\times3^{\frac{2+4}{3}} = 0.3\times3^2$

$= 0.3 \times 9 = 2.7$

17. (a); Expression $= \dfrac{2^n\times2^4-2\times2^n}{2\times2^n\times2^3}+\dfrac{1}{2^3}$

$= \dfrac{2^n(2^4-2)}{2^n(2\times2^3)}+\dfrac{1}{2^3} = \dfrac{16-2}{16}+\dfrac{1}{8} = \dfrac{7}{8}+\dfrac{1}{8} = 1$

18. (b); $10^{0.48} = x,\ 10^{0.7} = y$ and $x^z = y^2$

$\therefore\ (10^{0.48})^z = (10^{0.7})^2$

$\Rightarrow 10^{0.48z} = 10^{1.4} \Rightarrow 0.48z = 1.4$

$\Rightarrow z = \dfrac{1.4}{0.48} = \dfrac{140}{48} = \dfrac{35}{12} = 2\dfrac{11}{12}$

19. (a); Expression $= (27)^{-\frac{2}{3}}+\left[\left(2^{\frac{-2}{3}}\right)^{-\frac{5}{3}}\right]^{-\frac{9}{10}}$

$= (3^3)^{\frac{-2}{3}}+\left(2^{-\frac{2}{3}\times\frac{-5}{3}}\right)^{-\frac{9}{10}} = 3^{3\times-\frac{2}{3}}+2^{\frac{-2}{3}\times\frac{-5}{3}\times\frac{-9}{10}}$

$= 3^{-2}+2^{-1} = \dfrac{1}{3^2}+\dfrac{1}{2} = \dfrac{1}{9}+\dfrac{1}{2} = \dfrac{2+9}{18} = \dfrac{11}{18}$

20. (b); First term = $\sqrt{8+3\sqrt{7}}=\sqrt{\frac{(8+3\sqrt{7})\times 2}{2}}$

$$=\sqrt{\frac{16+6\sqrt{7}}{2}}=\sqrt{\frac{3^2+(\sqrt{7})^2+2\times 3\times\sqrt{7}}{2}}$$

$$=\sqrt{\frac{(3+\sqrt{7})^2}{2}}=\frac{3+\sqrt{7}}{\sqrt{2}}$$

Second term = $\sqrt{7+3\sqrt{5}}=\sqrt{\frac{(7+3\sqrt{5})\times 2}{2}}$

$$=\sqrt{\frac{14+6\sqrt{5}}{2}}=\sqrt{\frac{3^2+(\sqrt{5})^2+2\times 3\times\sqrt{5}}{2}}$$

$$=\sqrt{\frac{(3+\sqrt{5})^2}{2}}=\frac{3+\sqrt{5}}{\sqrt{2}}$$

$\therefore$ Expression = $\frac{3+\sqrt{7}}{\sqrt{2}}-\frac{3+\sqrt{5}}{\sqrt{2}}$

$$=\frac{3+\sqrt{7}-3-\sqrt{5}}{\sqrt{2}}=\frac{\sqrt{7}-\sqrt{5}}{\sqrt{2}}=\frac{\sqrt{14}-\sqrt{10}}{2}$$

21. (b); First term = $\sqrt{38+5\sqrt{3}}=\sqrt{\frac{(38+5\sqrt{3})\times 2}{2}}$

$$=\sqrt{\frac{76+10\sqrt{3}}{2}}=\sqrt{\frac{75+1+2\times 5\sqrt{3}\times 1}{2}}$$

$$=\sqrt{\frac{(5\sqrt{3}+1)^2}{2}}=\frac{5\sqrt{3}+1}{\sqrt{2}}$$

Second term = $\sqrt{3-\sqrt{5}}=\sqrt{(3-\sqrt{5})\times\frac{2}{2}}$

$$=\sqrt{\frac{6-2\sqrt{5}}{2}}=\sqrt{\frac{(\sqrt{5})^2+1-2\sqrt{5}}{2}}$$

$$=\sqrt{\frac{(\sqrt{5}-1)^2}{2}}=\frac{\sqrt{5}-1}{\sqrt{2}}$$

$\therefore$ Expression = $\frac{5\sqrt{3}+1}{\sqrt{2}}+\frac{\sqrt{5}-1}{\sqrt{2}}$

$$=\frac{5\sqrt{3}+1+\sqrt{5}-1}{\sqrt{2}}=\frac{5\sqrt{3}+\sqrt{5}}{\sqrt{2}}$$

$$=\frac{5\sqrt{3}+\sqrt{5}}{\sqrt{\ }}\times\frac{\sqrt{2}}{\sqrt{\ }}=\frac{5\sqrt{6}+\sqrt{10}}{2}$$

22. (a); First term = $\sqrt{4-\sqrt{7}}=\sqrt{\frac{(4-\sqrt{7})\times 2}{2}}$

$$=\sqrt{\frac{8-2\sqrt{7}}{2}}=\sqrt{\frac{(\sqrt{7})^2+1-2\sqrt{7}}{2}}$$

$$=\sqrt{\frac{(\sqrt{7}-1)^2}{2}}=\frac{\sqrt{7}-1}{\sqrt{2}}$$

Second term = $\sqrt{8+3\sqrt{7}}=\sqrt{\frac{(8+3\sqrt{7})\times 2}{2}}$

$$=\sqrt{\frac{16+6\sqrt{7}}{2}}=\sqrt{\frac{3^2+(\sqrt{7})^2+2\times 3\times\sqrt{7}}{2}}$$

$$=\sqrt{\frac{(3+\sqrt{7})^2}{2}}=\frac{3+\sqrt{7}}{\sqrt{2}}$$

$\therefore$ Expression = $\frac{\sqrt{7}-1}{\sqrt{2}}+\frac{3+\sqrt{7}}{\sqrt{2}}=\frac{2\sqrt{7}+2}{\sqrt{2}}$

$$=\frac{2\sqrt{14}+2\sqrt{2}}{2}=\sqrt{14}+\sqrt{2}=\sqrt{2}(\sqrt{7}+1)$$

23. (b); Denominator of expression

$$=\sqrt{10}+\sqrt{20}+\sqrt{40}-\sqrt{5}-\sqrt{80}$$

$$=\sqrt{10}+\sqrt{2^2\times 5}+\sqrt{2^2\times 10}-\sqrt{5}-\sqrt{2^4\times 5}$$

$$=\sqrt{10}+2\sqrt{5}+2\sqrt{10}-\sqrt{5}-4\sqrt{5}$$

$$=(1+2)\sqrt{10}+(2-1-4)\sqrt{5}$$

$$=3\sqrt{10}-3\sqrt{5}=3(\sqrt{10}-\sqrt{5})$$

$\therefore$ Expression = $\frac{15}{\sqrt{10}+\sqrt{20}+\sqrt{40}-\sqrt{5}-\sqrt{80}}$

$$=\frac{15}{3(\sqrt{10}-\sqrt{5})}=\frac{5}{\sqrt{10}-\sqrt{5}}\times\frac{\sqrt{10}+\sqrt{5}}{\sqrt{10}+\sqrt{5}}$$

$$=\frac{5(\sqrt{10}+\sqrt{5})}{10-5}=\sqrt{10}+\sqrt{5}$$

= 3.162 + 2.236 = 5.398

24. (b); Expression = $\frac{\left(2^{2n}-3.\frac{2^{2n}}{2^2}\right)\left(3^n-\frac{2.3^n}{3^2}\right)}{\frac{3^n}{3^4}(4^n.4^3-2^{2n})}$

$$=\frac{3^n.2^{2n}\left(1-\frac{3}{2^2}\right)\left(1-\frac{2}{3^2}\right)}{3^n.2^{2n}\left(\frac{2^6-1}{3^4}\right)}$$

$$=\frac{\left(1-\frac{3}{4}\right)\left(1-\frac{2}{9}\right)}{\frac{64-1}{81}}=\frac{\frac{1}{4}\times\frac{7}{9}}{\frac{63}{81}}=\frac{7}{36}\times\frac{81}{63}=\frac{1}{4}.$$

25. (b); Rationalising the denominator,

$$\frac{\sqrt{2}+\sqrt{3}}{3\sqrt{2}-2\sqrt{3}}=\frac{\sqrt{2}+\sqrt{3}}{3\sqrt{2}-2\sqrt{3}}\times\frac{3\sqrt{2}+2\sqrt{3}}{3\sqrt{2}+2\sqrt{3}}$$

$$=\frac{\sqrt{2}\times3\sqrt{2}+\sqrt{2}\times2\sqrt{3}+\sqrt{3}\times3\sqrt{2}+\sqrt{3}\times2\sqrt{3}}{(3\sqrt{2})^2-(2\sqrt{3})^2}$$

$$=\frac{3\sqrt{2\times2}+2\sqrt{2\times3}+3\sqrt{2\times3}+2\sqrt{3\times3}}{9\times2-4\times3}$$

$$=\frac{3\times2+2\sqrt{6}+3\sqrt{6}+2\times3}{18-12}$$

$$=\frac{12+5\sqrt{6}}{6}=\frac{12}{6}+\frac{5\sqrt{6}}{6}=2+\frac{5\sqrt{6}}{6}$$

$$\therefore\ \frac{\sqrt{2}+\sqrt{3}}{3\sqrt{2}-2\sqrt{3}}=a-b\sqrt{6}$$

$$\Rightarrow\ 2+\frac{5\sqrt{6}}{6}=a-b\sqrt{6}\ \Rightarrow\ a=2\ \text{ and }\ b=\frac{-5}{6}$$

$$\therefore\ (a+b)^2=\left(2-\frac{5}{6}\right)^2=\left(\frac{12-5}{6}\right)^2=\left(\frac{7}{6}\right)^2=\frac{49}{36}$$

26. (a); Rationalising the denominator,

$$\frac{7\sqrt{3}}{\sqrt{10}+\sqrt{3}}-\frac{2\sqrt{5}}{\sqrt{6}+\sqrt{5}}-\frac{3\sqrt{2}}{\sqrt{15}+3\sqrt{2}}$$

$$=\frac{7\sqrt{3}}{\sqrt{10}+\sqrt{3}}\times\frac{\sqrt{10}-\sqrt{3}}{\sqrt{10}-\sqrt{3}}-\frac{2\sqrt{5}}{\sqrt{6}+\sqrt{5}}\times\frac{\sqrt{6}-\sqrt{5}}{\sqrt{6}-\sqrt{5}}-\frac{3\sqrt{2}}{\sqrt{15}+3\sqrt{2}}\times\frac{\sqrt{15}-3\sqrt{2}}{\sqrt{15}-3\sqrt{2}}$$

$$=\frac{7\sqrt{3}(\sqrt{10}-\sqrt{3})}{(\sqrt{10})^2-(\sqrt{3})^2}-\frac{2\sqrt{5}(\sqrt{6}-\sqrt{5})}{(\sqrt{6})^2-(\sqrt{5})^2}-\frac{3\sqrt{2}(\sqrt{15}-3\sqrt{2})}{(\sqrt{15})^2-(3\sqrt{2})^2}$$

$$=\frac{7\sqrt{3}(\sqrt{10}-\sqrt{3})}{10-3}-\frac{2\sqrt{5}(\sqrt{6}-\sqrt{5})}{6-5}-\frac{3\sqrt{2}(\sqrt{15}-3\sqrt{2})}{15-9\times2}$$

$$=\sqrt{3}(\sqrt{10}-\sqrt{3})-2\sqrt{5}(\sqrt{6}-\sqrt{5})+\sqrt{2}(\sqrt{15}-3\sqrt{2})$$

$$=\sqrt{30}-3-2\sqrt{30}+2\times5+\sqrt{30}-3\times2$$

$$=-3+10-6+\sqrt{30}-2\sqrt{30}+\sqrt{30}$$

$$=1+(1-2+1)\sqrt{30}=1$$

27. (b); $\dfrac{\sqrt{5}-2}{\sqrt{5}+2}-\dfrac{\sqrt{5}+2}{\sqrt{5}-2}=\dfrac{(\sqrt{5}-2)^2-(\sqrt{5}+2)^2}{(\sqrt{5}+2)(\sqrt{5}-2)}$

$$=\frac{\left[(\sqrt{5})^2+2^2-2\times\sqrt{5}\times2\right]-\left[(\sqrt{5})^2+2^2+2\times\sqrt{5}\times2\right]}{(\sqrt{5})^2-(2)^2}$$

$$=\frac{(5+4-4\sqrt{5})-(5+4+4\sqrt{5})}{5-4}$$

$$=9-4\sqrt{5}-9-4\sqrt{5}=-8\sqrt{5}$$

28. (c); Rationalising the denominator,

$$\frac{1}{3-\sqrt{8}}\times\frac{3+\sqrt{8}}{3+\sqrt{8}}-\frac{1}{\sqrt{8}-\sqrt{7}}\times\frac{\sqrt{8}+\sqrt{7}}{\sqrt{8}+\sqrt{7}}+\frac{1}{\sqrt{7}-\sqrt{6}}\times\frac{\sqrt{7}+\sqrt{6}}{\sqrt{7}+\sqrt{6}}-\frac{1}{\sqrt{6}-\sqrt{5}}\times\frac{\sqrt{6}+\sqrt{5}}{\sqrt{6}+\sqrt{5}}+\frac{1}{\sqrt{5}-2}\times\frac{\sqrt{5}+2}{\sqrt{5}+2}$$

$$=\frac{3+\sqrt{8}}{3^2-(\sqrt{8})^2}-\frac{\sqrt{8}+\sqrt{7}}{(\sqrt{8})^2-(\sqrt{7})^2}+\frac{\sqrt{7}+\sqrt{6}}{(\sqrt{7})^2-(\sqrt{6})^2}-\frac{\sqrt{6}+\sqrt{5}}{(\sqrt{6})^2-(\sqrt{5})^2}+\frac{\sqrt{5}+2}{(\sqrt{5})^2-(2)^2}$$

$$=\frac{3+\sqrt{8}}{9-8}-\frac{\sqrt{8}+\sqrt{7}}{8-7}+\frac{\sqrt{7}+\sqrt{6}}{7-6}-\frac{\sqrt{6}+\sqrt{5}}{6-5}+\frac{\sqrt{5}+2}{5-4}$$

$$=(3+\sqrt{8})-(\sqrt{8}+\sqrt{7})+(\sqrt{7}+\sqrt{6})-(\sqrt{6}+\sqrt{5})+(\sqrt{5}+2)$$

$$=3+\sqrt{8}-\sqrt{8}-\sqrt{7}+\sqrt{7}+\sqrt{6}-\sqrt{6}-\sqrt{5}+\sqrt{5}+2$$
$$=3+2=5$$

29. (d); Expression $=(9)^{-3}\times\dfrac{(16)^{\frac{1}{4}}}{(6)^{-2}}\times\left(\dfrac{1}{27}\right)^{-\frac{4}{3}}$

$$=(3^2)^{-3}\times\frac{(2^4)^{\frac{1}{4}}}{(2\times3)^{-2}}\times\left(\frac{1}{3^3}\right)^{-\frac{4}{3}}$$

$$=3^{-6}\times\frac{2}{2^{-2}\times3^{-2}}\times(3^3)^{\frac{4}{3}}\qquad\left[\because\left(\frac{a}{b}\right)^{-n}=\left(\frac{b}{a}\right)^{n}\right]$$

$$=3^{-6}\times2\times3^4\times2^2\times3^2\qquad\left[\because a^n=\frac{1}{a^{-n}}\right]$$

$= 3^{-6} \times 3^4 \times 3^2 \times 2^2 \times 2$

$= 3^{-6+6} \times 2^{2+1}$ $\quad [\because a^m.a^n = a^{m+n}]$

$= 3^0 \times 2^3 = 1 \times 8 = 8$

30. **(a);** Let $x = \sqrt{7\sqrt{7\sqrt{7\ldots\ldots\ldots\infty}}}$

Squaring both sides,

$x^2 = 7\sqrt{7\sqrt{7\sqrt{7\ldots\ldots\ldots\infty}}}$

$\Rightarrow x^2 = 7x \quad \Rightarrow x^2 - 7x = 0$

$\Rightarrow x(x-7) = 0 \quad \Rightarrow \quad x = 0$ or, $x = 7$

But $x \neq 0$

$\therefore$ Required answer $= 7$

Difficult

1. **(b);** Given $x = \frac{\sqrt{3}}{2}$

$$\frac{\sqrt{1+x}}{1+\sqrt{1+x}} \times \frac{1-\sqrt{1+x}}{1-\sqrt{1+x}} + \frac{\sqrt{1-x}}{1-\sqrt{1-x}} \times \frac{1+\sqrt{1-x}}{1+\sqrt{1-x}}$$

$$= \frac{\sqrt{1+x}-1-x}{1-1-x} + \frac{\sqrt{1-x}+1-x}{1-1+x}$$

$$= \frac{\sqrt{1-x}+1-x}{x} - \frac{\sqrt{1+x}-1-x}{x}$$

$$= \frac{\sqrt{1-x}+1-x-\sqrt{1+x}+1+x}{x}$$

$$= \frac{2+\sqrt{1-x}-\sqrt{1+x}}{x} = \frac{2+\sqrt{1-\frac{\sqrt{3}}{2}}-\sqrt{1+\frac{\sqrt{3}}{2}}}{\frac{\sqrt{3}}{2}}$$

$$= \frac{2+\sqrt{\frac{2-\sqrt{3}}{2}}-\sqrt{\frac{2+\sqrt{3}}{2}}}{\frac{\sqrt{3}}{2}}$$

$$= \frac{2+\frac{\sqrt{4-2\sqrt{3}}}{2}-\frac{\sqrt{4+2\sqrt{3}}}{2}}{\frac{\sqrt{3}}{2}}$$

$$= \frac{4+\sqrt{3}-1-\sqrt{3}-1}{\sqrt{3}} = \frac{2}{\sqrt{3}}$$

2. **(b);** $x = \frac{2\sqrt{24}}{\sqrt{3}+\sqrt{2}}$

$$\Rightarrow x = \frac{2\sqrt{3\times 8}}{\sqrt{3}+\sqrt{2}} = \frac{2\sqrt{3}\times\sqrt{8}}{\sqrt{3}+\sqrt{2}}$$

$$\Rightarrow \frac{x}{\sqrt{8}} = \frac{2\sqrt{3}}{\sqrt{3}+\sqrt{2}}$$

$$\Rightarrow \frac{x+\sqrt{8}}{x-\sqrt{8}} = \frac{2\sqrt{3}+\sqrt{3}+\sqrt{2}}{2\sqrt{3}-\sqrt{3}-\sqrt{2}}$$

(By componendo and dividendo)

$$= \frac{3\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$$

Again,

$$\frac{x}{\sqrt{12}} = \frac{2\sqrt{2}}{\sqrt{3}+\sqrt{2}} \quad \Rightarrow \quad \frac{x+\sqrt{12}}{x-\sqrt{12}}$$

$$= \frac{2\sqrt{2}+\sqrt{3}+\sqrt{2}}{2\sqrt{2}-\sqrt{3}-\sqrt{2}} = \frac{\sqrt{3}+3\sqrt{2}}{\sqrt{2}-\sqrt{3}}$$

$$\therefore \frac{x+\sqrt{8}}{x-\sqrt{8}} + \frac{x+\sqrt{12}}{x-\sqrt{12}} = \frac{3\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}} + \frac{\sqrt{3}+3\sqrt{2}}{\sqrt{2}-\sqrt{3}}$$

$$= \frac{3\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}} - \frac{\sqrt{3}+3\sqrt{2}}{\sqrt{3}-\sqrt{2}} = \frac{3\sqrt{3}+\sqrt{2}-\sqrt{3}-3\sqrt{2}}{\sqrt{3}-\sqrt{2}}$$

$$= \frac{2\sqrt{3}-2\sqrt{2}}{\sqrt{3}-\sqrt{2}} = \frac{2(\sqrt{3}-\sqrt{2})}{\sqrt{3}-\sqrt{2}} = 2$$

3. **(d);** $x = \frac{2\sqrt{3}\times\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

$$\Rightarrow \frac{x}{\sqrt{2}} = \frac{2\sqrt{3}}{\sqrt{3}+\sqrt{2}}$$

$$\Rightarrow \frac{x+\sqrt{2}}{x-\sqrt{2}} = \frac{2\sqrt{3}+\sqrt{3}+\sqrt{2}}{2\sqrt{3}-\sqrt{3}-\sqrt{2}} = \frac{3\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$$

(By componendo and dividendo)

Similarly,

$$\frac{x}{\sqrt{3}} = \frac{2\sqrt{2}}{\sqrt{3}+\sqrt{2}}$$

$$\Rightarrow \frac{x+\sqrt{3}}{x-\sqrt{3}} = \frac{2\sqrt{2}+\sqrt{3}+\sqrt{2}}{2\sqrt{2}-\sqrt{3}-\sqrt{2}} = \frac{\sqrt{3}+3\sqrt{2}}{\sqrt{2}-\sqrt{3}}$$

$\therefore$ Expression

$$= \frac{x+\sqrt{2}}{x-\sqrt{2}} + \frac{x+\sqrt{3}}{x-\sqrt{3}} = \frac{3\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}} + \frac{\sqrt{3}+3\sqrt{2}}{\sqrt{2}-\sqrt{3}}$$

$$= \frac{3\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}} - \frac{\sqrt{3}+3\sqrt{2}}{\sqrt{3}-\sqrt{2}} = \frac{3\sqrt{3}+\sqrt{2}-\sqrt{3}-3\sqrt{2}}{\sqrt{3}-\sqrt{2}}$$

$$= \frac{2(\sqrt{3}-\sqrt{2})}{\sqrt{3}-\sqrt{2}} = 2$$

4. **(c);** $x = \frac{\sqrt{3}}{2}$

$$\therefore \sqrt{1+x} = \sqrt{1+\frac{\sqrt{3}}{2}}$$

$$= \sqrt{\frac{2+\sqrt{3}}{2}} = \sqrt{\frac{4+2\sqrt{3}}{4}} = \sqrt{\frac{(\sqrt{3}+1)^2}{4}} = \frac{\sqrt{3}+1}{2}$$

$$\therefore \sqrt{1-x} = \frac{\sqrt{3}-1}{2}$$

$$\therefore \sqrt{1+x} + \sqrt{1-x}$$

$$= \frac{\sqrt{3}+1}{2} + \frac{\sqrt{3}-1}{2} = \frac{\sqrt{3}+1+\sqrt{3}-1}{2} = \sqrt{3}$$

5. **(a);** First part = $\sqrt{11-2\sqrt{30}} = \sqrt{11-2\sqrt{6\times5}}$

$$= \sqrt{11-2\times\sqrt{6}\times\sqrt{5}} = \sqrt{6+5-2\times\sqrt{6}\times\sqrt{5}}$$

$$= \sqrt{(\sqrt{6})^2+(\sqrt{5})^2-2\times\sqrt{6}\times\sqrt{5}}$$

$$= \sqrt{(\sqrt{6}-\sqrt{5})^2} = \sqrt{6}-\sqrt{5}$$

Second part = $\sqrt{7-2\sqrt{10}}$

$$= \sqrt{7-2\sqrt{5\times2}} = \sqrt{7-2\sqrt{5}\times\sqrt{2}}$$

$$= \sqrt{5+2-2\sqrt{5}\times\sqrt{2}}$$

$$= \sqrt{(\sqrt{5})^2+(\sqrt{2})^2-2\times\sqrt{5}\times\sqrt{2}}$$

$$= \sqrt{(\sqrt{5}-\sqrt{2})^2} = \sqrt{5}-\sqrt{2}$$

Third part = $\frac{4}{\sqrt{6}+\sqrt{2}}$

Rationalising the denominator,

$$= \frac{4}{(\sqrt{6}+\sqrt{2})} \times \frac{(\sqrt{6}-\sqrt{2})}{(\sqrt{6}-\sqrt{2})}$$

$$= \frac{4(\sqrt{6}-\sqrt{2})}{(\sqrt{6})^2-(\sqrt{2})^2} = \frac{4(\sqrt{6}-\sqrt{2})}{6-2} = \sqrt{6}-\sqrt{2}$$

$\therefore$ Expression

$$= (\sqrt{6}-\sqrt{5}) + (\sqrt{5}-\sqrt{2}) - (\sqrt{6}-\sqrt{2})$$

$$= \sqrt{6}-\sqrt{5}+\sqrt{5}-\sqrt{2}-\sqrt{6}+\sqrt{2} = 0$$

6. **(c);** First part = $\sqrt{\frac{(2.4)^6+9(5.76)+6(2.4)^4}{(2.4)^4+6(5.76)+9}}$

$$= \sqrt{\frac{(2.4)^6+9(2.4)^2+6(2.4)^4}{(2.4)^4+6(2.4)^2+9}}$$

$$= \sqrt{\frac{(2.4)^2\left[(2.4)^4+9+6(2.4)^2\right]}{(2.4)^4+6(2.4)^2+9}} = \sqrt{(2.4)^2} = 2.4$$

Second part = $\frac{\left[(3^{-2})^{-5}\right]^{\frac{1}{5}} + \left[(4^{-3})^{-6}\right]^{\frac{1}{6}} - (3^{-4})^{-\frac{1}{2}}}{\left[(2^{-3})^{-4}\right]^{\frac{1}{4}}}$

$$= \frac{(3)^{(-2)\times(-5)\times\left(\frac{1}{5}\right)} + (4)^{(-3)\times(-6)\times\frac{1}{6}} - (3)^{(-4)\times\left(-\frac{1}{2}\right)}}{(2)^{(-3)\times(-4)\times\left(\frac{1}{4}\right)}}$$

$$\left[\because \left[(a^x)^y\right]^z = a^{xyz}\right]$$

$$= \frac{3^2+4^3-3^2}{2^3} = \frac{9+64-9}{8} = \frac{64}{8} = 8$$

$\therefore$ Expression = 2.4 + 8 = 10.4

7. **(d);** $\left(\frac{x^a}{x^{-b}}\right)^{a^2+b^2-ab} \times \left(\frac{x^b}{x^{-c}}\right)^{b^2+c^2-bc} \times \left(\frac{x^c}{x^{-a}}\right)^{c^2+a^2-ca}$

$$= \left(x^{a+b}\right)^{a^2+b^2-ab} \times \left(x^{b+c}\right)^{b^2+c^2-bc} \times \left(x^{c+a}\right)^{c^2+a^2-ca}$$

$$= x^{(a+b)(a^2+b^2-ab)} \times x^{(b+c)(b^2+c^2-bc)} \times x^{(c+a)(c^2+a^2-ca)}$$

$$= x^{a^3+b^3} \times x^{b^3+c^3} \times x^{c^3+a^3}$$

$$= x^{a^3+b^3+b^3+c^3+c^3+a^3} = x^{2(a^3+b^3+c^3)}$$

8. **(b);** It is given $2^x = 4^y = 8^z = k$ (Let)

$$\Rightarrow 2 = k^{\frac{1}{x}}, 4 = k^{\frac{1}{y}}, 8 = k^{\frac{1}{z}}$$

Now, $4 \times 8 = 32 = 2^5$

$$\Rightarrow k^{\frac{1}{y}} \times k^{\frac{1}{z}} = \left(k^{\frac{1}{x}}\right)^5 \qquad \Rightarrow k^{\frac{1}{y}+\frac{1}{z}} = k^{\frac{5}{x}}$$

$$\Rightarrow \frac{1}{y} + \frac{1}{z} = \frac{5}{x}$$

Now, $\frac{1}{2x}+\frac{1}{4y}+\frac{1}{4z}=4$

$\Rightarrow \frac{1}{2x}+\frac{1}{4}\left(\frac{1}{y}+\frac{1}{z}\right)=4 \quad \Rightarrow \frac{1}{2x}+\frac{1}{4}\left(\frac{5}{x}\right)=4$

$\Rightarrow \frac{1}{2x}+\frac{5}{4x}=4 \quad \Rightarrow \frac{2+5}{4x}=4$

$\Rightarrow \frac{7}{4x}=4 \Rightarrow 16x=7 \Rightarrow x=\frac{7}{16}$

9. (c); Let $x=\frac{\sqrt{\sqrt{5}+2}+\sqrt{\sqrt{5}-2}}{\sqrt{\sqrt{5}+1}}$

On squaring both sides.

$\Rightarrow x^2=\left[\frac{\sqrt{\sqrt{5}+2}+\sqrt{\sqrt{5}-2}}{\sqrt{\sqrt{5}+1}}\right]^2$

$=\frac{\left[\sqrt{\sqrt{5}+2}+\sqrt{\sqrt{5}-2}\right]^2}{\left[\sqrt{\sqrt{5}+1}\right]^2}$

$=\frac{\sqrt{5}+2+\sqrt{5}-2+2\sqrt{\sqrt{5}+2}\cdot\sqrt{\sqrt{5}-2}}{\sqrt{5}+1}$

$=\frac{2\sqrt{5}+2\sqrt{(\sqrt{5})^2-(2)^2}}{\sqrt{5}+1}$

$=\frac{2\sqrt{5}+2}{\sqrt{5}+1}=\frac{2(\sqrt{5}+1)}{\sqrt{5}+1}=2 \quad \therefore x=\sqrt{2}$

Again, $\sqrt{3-2\sqrt{2}}=\sqrt{2+1-2\times\sqrt{2}\times 1}$

$=\sqrt{(\sqrt{2})^2+(1)^2-2\times\sqrt{2}\times 1}$

$=\sqrt{(\sqrt{2}-1)^2}=\sqrt{2}-1$

$\therefore \frac{\sqrt{\sqrt{5}+2}+\sqrt{\sqrt{5}-2}}{\sqrt{\sqrt{5}+1}}-\sqrt{3-2\sqrt{2}}$

$=\sqrt{2}-(\sqrt{2}-1)=1$

10. (d); First term = $\sqrt{11+4\sqrt{7}}$

$=\sqrt{2^2+(\sqrt{7})^2+2\times 2\times\sqrt{7}}=\sqrt{(2+\sqrt{7})^2}=2+\sqrt{7}$

Second term

$=\sqrt{11-4\sqrt{7}}=\sqrt{(2)^2+(\sqrt{7})^2-2\times 2\times\sqrt{7}}$

$=\sqrt{(\sqrt{7}-2)^2}=\sqrt{7}-2$

Third term = $\frac{\sqrt{8}}{\sqrt{33+10\sqrt{8}}-\sqrt{33-10\sqrt{8}}}$

$=\frac{\sqrt{8}}{\sqrt{(5)^2+(\sqrt{8})^2+2\times 5\times\sqrt{8}}-\sqrt{(5)^2+(\sqrt{8})^2-2\times 5\times\sqrt{8}}}$

$=\frac{\sqrt{8}}{\sqrt{(5+\sqrt{8})^2}-\sqrt{(5-\sqrt{8})^2}}$

$=\frac{\sqrt{8}}{5+\sqrt{8}-5+\sqrt{8}}=\frac{\sqrt{8}}{2\sqrt{8}}=\frac{1}{2}$

$\therefore$ Expression = $2+\sqrt{7}-\sqrt{7}+2+\frac{1}{2}=4\frac{1}{2}$

11. (b); Expression

$=\frac{26-15\sqrt{3}}{\left[5\sqrt{2}-\sqrt{38+5\sqrt{3}}\right]^2}+\frac{\sqrt{10}+\sqrt{18}}{\sqrt{8}+\sqrt{(3-\sqrt{5})}}$

Now, $\sqrt{38+5\sqrt{3}}=\sqrt{(38+5\sqrt{3})\times\frac{2}{2}}$

$=\sqrt{\frac{76+2(5\sqrt{3})}{2}}=\sqrt{\frac{75+1+2(5\sqrt{3})}{2}}$

$=\sqrt{\frac{(5\sqrt{3})^2+(1)^2+2(5\sqrt{3})\times 1}{2}}$

$=\sqrt{\frac{(5\sqrt{3}+1)^2}{2}}=\frac{5\sqrt{3}+1}{\sqrt{2}}$

$\therefore \left(5\sqrt{2}-\sqrt{38+5\sqrt{3}}\right)^2=\left(5\sqrt{2}-\frac{5\sqrt{3}+1}{\sqrt{2}}\right)^2$

$=\left(\frac{10-5\sqrt{3}-1}{\sqrt{2}}\right)^2=\frac{(9-5\sqrt{3})^2}{2}$

and $\sqrt{(3-\sqrt{5})}=\sqrt{(3-\sqrt{5})\times\frac{2}{2}}$

$=\sqrt{\frac{6-2\sqrt{5}}{2}}=\sqrt{\frac{5+1-2(\sqrt{5})\times 1}{2}}$

$=\sqrt{\frac{(\sqrt{5})^2+(1)^2-2\sqrt{5}}{2}}=\sqrt{\frac{(\sqrt{5}-1)^2}{2}}=\frac{\sqrt{5}-1}{\sqrt{2}}$

$$\therefore \frac{26-15\sqrt{3}}{\left(5\sqrt{2}-\frac{5\sqrt{3}+1}{\sqrt{2}}\right)^2}+\frac{\sqrt{2}(\sqrt{5}+3)}{\sqrt{8}+\frac{\sqrt{5}-1}{\sqrt{2}}}$$

$$=\frac{2(26-15\sqrt{3})}{(9-5\sqrt{3})^2}+\frac{\sqrt{2}\sqrt{2}(\sqrt{5}+3)}{4+\sqrt{5}-1}$$

$$=\frac{52-30\sqrt{3}}{156-90\sqrt{3}}+\frac{2(3+\sqrt{5})}{(3+\sqrt{5})}$$

$$=\frac{52-30\sqrt{3}}{3(52-30\sqrt{3})}+2=\frac{1}{3}+2=2\frac{1}{3}$$

12. (c); First term = $(28-10\sqrt{3})^{\frac{1}{2}}$

$$=(25+3-10\sqrt{3})^{\frac{1}{2}}=\left[(5)^2+(\sqrt{3})^2-2\times5\times\sqrt{3}\right]^{\frac{1}{2}}$$

$$=\left[(5-\sqrt{3})^2\right]^{\frac{1}{2}}=(5-\sqrt{3})^{2\times\frac{1}{2}}=(5-\sqrt{3})$$

Second term = $(7+4\sqrt{3})^{-\frac{1}{2}}=(4+3+4\sqrt{3})^{-\frac{1}{2}}$

$$=\left[(2)^2+(\sqrt{3})^2+2\times2\times\sqrt{3}\right]^{-\frac{1}{2}}$$

$$=\left[(2+\sqrt{3})^2\right]^{-\frac{1}{2}}=(2+\sqrt{3})^{2\times-\frac{1}{2}}$$

$$=(2+\sqrt{3})^{-1}=\frac{1}{(2+\sqrt{3})}$$

$$=\frac{1}{2+\sqrt{3}}\times\frac{2-\sqrt{3}}{2-\sqrt{3}}=\frac{2-\sqrt{3}}{4-3}=2-\sqrt{3}$$

Third term = $\frac{\sqrt{7}}{\sqrt{16+6\sqrt{7}}-\sqrt{16-6\sqrt{7}}}$

$$=\frac{\sqrt{7}}{\sqrt{9+7+2\times3\times\sqrt{7}}-\sqrt{9+7-2\times3\times\sqrt{7}}}$$

$$=\frac{\sqrt{7}}{\sqrt{(3+\sqrt{7})^2}-\sqrt{(3-\sqrt{7})^2}}$$

$$=\frac{\sqrt{7}}{(3+\sqrt{7})-(3-\sqrt{7})}=\frac{\sqrt{7}}{3+\sqrt{7}-3+\sqrt{7}}$$

(Taking square root)

$$=\frac{\sqrt{7}}{2\sqrt{7}}=\frac{1}{2}$$

$\therefore$ Expression = $(28-10\sqrt{3})^{\frac{1}{2}}-(7+4\sqrt{3})^{-\frac{1}{2}}$

$$+\frac{\sqrt{7}}{\sqrt{16+6\sqrt{7}}-\sqrt{16-6\sqrt{7}}}$$

$$=5-\sqrt{3}-(2-\sqrt{3})+\frac{1}{2}$$

$$=5-\sqrt{3}-2+\sqrt{3}+\frac{1}{2}=3+\frac{1}{2}=3\frac{1}{2}$$

13. (d); $\frac{4\sqrt{3}}{2-\sqrt{2}}-\frac{30}{4\sqrt{3}-\sqrt{18}}-\frac{\sqrt{18}}{3-2\sqrt{3}}$

Rationalising the denominators by corresponding conjugates

$$=\frac{4\sqrt{3}(2+\sqrt{2})}{(2-\sqrt{2})(2+\sqrt{2})}$$

$$-\frac{30(4\sqrt{3}+\sqrt{18})}{(4\sqrt{3}-\sqrt{18})(4\sqrt{3}+\sqrt{18})}-\frac{\sqrt{18}(3+2\sqrt{3})}{(3-2\sqrt{3})(3+2\sqrt{3})}$$

$$=\frac{4\sqrt{3}(2+\sqrt{2})}{(2)^2-(\sqrt{2})^2}-\frac{30(4\sqrt{3}+\sqrt{18})}{(4\sqrt{3})^2-(\sqrt{18})^2}-\frac{\sqrt{18}(3+2\sqrt{3})}{(3)^2-(2\sqrt{3})^2}$$

$[(a-b)(a+b)=a^2-b^2]$

$$=\frac{4\sqrt{3}(2+\sqrt{2})}{4-2}-\frac{30(4\sqrt{3}+\sqrt{18})}{48-18}-\frac{\sqrt{18}(3+2\sqrt{3})}{9-12}$$

$$=\frac{4\sqrt{3}(2+\sqrt{2})}{2}-\frac{30(4\sqrt{3}+\sqrt{18})}{30}-\frac{3\sqrt{2}(3+2\sqrt{3})}{-3}$$

$$=2\sqrt{3}(2+\sqrt{2})-(4\sqrt{3}+\sqrt{18})+\sqrt{2}(3+2\sqrt{3})$$

$$=4\sqrt{3}+2\sqrt{6}-4\sqrt{3}-3\sqrt{2}+3\sqrt{2}+2\sqrt{6}$$

$$=2\sqrt{6}+2\sqrt{6}=4\sqrt{6}$$

14. (c); First term = $\sqrt{\frac{(12.12)^2-(8.12)^2}{(0.25)^2+(0.25)(19.99)}}$

$$=\sqrt{\frac{(12.12+8.12)(12.12-8.12)}{0.25(0.25+19.99)}}$$

$\left[\because(a^2-b^2)=(a+b)(a-b)\right]$

$$=\sqrt{\frac{20.24\times4}{0.25\times20.24}}=\sqrt{\frac{4}{0.25}}=\sqrt{\frac{4\times100}{25}}$$

$$= \sqrt{\frac{400}{25}} = \sqrt{16} = 4$$

$$\text{Second term} = \frac{\left[\left(8^{\frac{-3}{4}}\right)^{\frac{5}{2}}\right]^{\frac{8}{15}} \times 16^{\frac{3}{4}}}{\sqrt[3]{\left[\left\{(128)^{-5}\right\}^{\frac{3}{7}}\right]^{-\frac{1}{5}}}}$$

$$= \frac{\left[\left\{(2^3)^{-\frac{3}{4}}\right\}^{\frac{5}{2}}\right]^{\frac{8}{15}} \times (2^4)^{\frac{3}{4}}}{\sqrt[3]{\left[\left\{(2^7)^{-5}\right\}^{\frac{3}{7}}\right]^{-\frac{1}{5}}}} \left[\because 8 = 2^3, 16 = 2^4, 128 = 2^7\right]$$

$$= \frac{2^{3\times\frac{-3}{4}\times\frac{5}{2}\times\frac{8}{15}} \times 2^{4\times\frac{3}{4}}}{2^{7\times(-5)\times\frac{3}{7}\times\left(-\frac{1}{5}\right)\times\frac{1}{3}}}$$

$$\left[\because \left[\left\{(a^m)^n\right\}^p\right]^q = a^{mnpq} \text{ and } \sqrt[3]{x} = (x)^{\frac{1}{3}}\right]$$

$$= \frac{2^{-3} \times 2^3}{2} = \frac{2^{-3+3}}{2} = \frac{2^0}{2} = \frac{1}{2}$$

$\therefore$ Expression = First term + second term

$$= 4 + \frac{1}{2} = 4\frac{1}{2}$$

15. (d); First term $= \dfrac{\sqrt[6]{2}\left[(625)^{\frac{3}{5}} \times (1024)^{-\frac{6}{5}} \div (25)^{\frac{3}{5}}\right]^{\frac{1}{2}}}{\left(\sqrt[3]{128}\right)^{-\frac{5}{2}} \times (125)^{\frac{1}{5}}}$

$$= \frac{2^{\frac{1}{6}}\left[(5^4)^{\frac{3}{5}} \times (2^{10})^{-\frac{6}{5}} \div (5^2)^{\frac{3}{5}}\right]^{\frac{1}{2}}}{\left\{(128)^{\frac{1}{3}}\right\}^{-\frac{5}{2}} \times (5^3)^{\frac{1}{5}}}$$

$$= \frac{2^{\frac{1}{6}}\left[5^{4\times\frac{3}{5}} \times 2^{10\times\frac{-6}{5}} \div 5^{2\times\frac{3}{5}}\right]^{\frac{1}{2}}}{\left\{(2^7)^{\frac{1}{3}}\right\}^{-\frac{5}{2}} \times 5^{3\times\frac{1}{5}}}$$

$$= \frac{2^{\frac{1}{6}}\left[5^{\frac{12}{5}} \times 2^{-12} \div 5^{\frac{6}{5}}\right]^{\frac{1}{2}}}{2^{\left(7\times\frac{1}{3}\times\frac{-5}{2}\right)} \times 5^{\frac{3}{5}}} = \frac{2^{\frac{1}{6}}\left[5^{\frac{6}{5}} \times 2^{-12}\right]^{\frac{1}{2}}}{2^{\frac{-35}{6}} \times 5^{\frac{3}{5}}}$$

$$= \frac{2^{\frac{1}{6}} \times 5^{\frac{3}{5}} \times 2^{-6}}{2^{\frac{-35}{6}} \times 5^{\frac{3}{5}}} = 2^{\left(\frac{1}{6}-6+\frac{35}{6}\right)} \times 5^{\frac{3}{5}-\frac{3}{5}}$$

$$= 2^{\left(\frac{1-36+35}{6}\right)} \times 5^0 = 2^{\frac{36-36}{6}} \times 5^0$$

$$= 2^0 \times 5^0 = 1 \times 1 = 1$$

$$\text{Second term} = \frac{(10^3)^2 \div \left(10^{3^2}\right)}{(10^2)^3 \div 10^{2^3}} = \frac{10^{2\times3} \div 10^9}{10^{2\times3} \div 10^8}$$

$$= \frac{10^6 \div 10^9}{10^6 \div 10^8} = \frac{10^{(6-9)}}{10^{(6-8)}}$$

$$= \frac{10^{-3}}{10^{-2}} = 10^{(-3+2)} = 10^{-1} = \frac{1}{10}$$

$\therefore$ Expression

$$= \frac{\sqrt[6]{2}\left[(625)^{\frac{3}{5}} \times (1024)^{-\frac{6}{5}} \div (25)^{\frac{3}{5}}\right]^{\frac{1}{2}}}{\left(\sqrt[3]{128}\right)^{-\frac{5}{2}} \times (125)^{\frac{1}{5}}} + \frac{(10^3)^2 \div \left(10^{3^2}\right)}{(10^2)^3 \div 10^{2^3}}$$

$$= 1 + \frac{1}{10} = \frac{10+1}{10} = 1.1$$

16. (a); First term $= \left(18 - 8\sqrt{2}\right)^{\frac{1}{2}}$

$$= \left(16 + 2 - 8\sqrt{2}\right)^{\frac{1}{2}} = \left\{(4)^2 + \left(\sqrt{2}\right)^2 - 2 \times 4 \times \sqrt{2}\right\}^{\frac{1}{2}}$$

$$= \left(4 - \sqrt{2}\right)^{2\times\frac{1}{2}} = 4 - \sqrt{2}$$

$$\text{Second term} = \left(6 - 4\sqrt{2}\right)^{-\frac{1}{2}} = \left(4 + 2 - 4\sqrt{2}\right)^{-\frac{1}{2}}$$

$$= \left\{2^2 + \left(\sqrt{2}\right)^2 - 2 \times 2 \times \sqrt{2}\right\}^{-\frac{1}{2}}$$

$$= \left(2 - \sqrt{2}\right)^{2\times\frac{-1}{2}} = \left(2 - \sqrt{2}\right)^{-1} = \frac{1}{2 - \sqrt{2}}$$

$$= \frac{1}{2-\sqrt{2}} \times \frac{2+\sqrt{2}}{2+\sqrt{2}} = \frac{2+\sqrt{2}}{2}$$

$$\text{Third term} = \frac{\sqrt{3}}{\sqrt{19+8\sqrt{3}} - \sqrt{19-8\sqrt{3}}}$$

$$= \frac{\sqrt{3}}{\sqrt{16+3+8\sqrt{3}} - \sqrt{16+3-8\sqrt{3}}}$$

$$= \frac{\sqrt{3}}{\sqrt{(4)^2 + \left(\sqrt{3}\right)^2 + 2\times4\times\sqrt{3}} - \sqrt{(4)^2 + \left(\sqrt{3}\right)^2 - 2\times4\times\sqrt{3}}}$$

$$= \frac{\sqrt{3}}{\sqrt{\left(4+\sqrt{3}\right)^2} - \sqrt{\left(4-\sqrt{3}\right)^2}}$$

$$= \frac{\sqrt{3}}{4+\sqrt{3}-4+\sqrt{3}} = \frac{\sqrt{3}}{2\sqrt{3}} = \frac{1}{2}$$

$\therefore$ Expression $= 4-\sqrt{2}+\frac{2+\sqrt{2}}{2}+\frac{1}{2}$

$= \frac{8-2\sqrt{2}+2+\sqrt{2}+1}{2} = \frac{11-\sqrt{2}}{2}$

17. (d); First term $= \sqrt{7+4\sqrt{3}} = \sqrt{2^2+(\sqrt{3})^2+2\times2\times\sqrt{3}}$

$= \sqrt{(2+\sqrt{3})^2} = 2+\sqrt{3}$

Second term

$= \sqrt{28+10\sqrt{3}} = \sqrt{(5)^2+(\sqrt{3})^2+2\times5\times\sqrt{3}}$

$= \sqrt{(5+\sqrt{3})^2} = 5+\sqrt{3}$

Third term $= \frac{\sqrt{11}}{\sqrt{20+6\sqrt{11}}+\sqrt{20-6\sqrt{11}}}$

$= \frac{\sqrt{11}}{\sqrt{3^2+(\sqrt{11})^2+2\times3\times\sqrt{11}}+\sqrt{3^2+(\sqrt{11})^2-2\times3\times\sqrt{11}}}$

$= \frac{\sqrt{11}}{\sqrt{(\sqrt{11}+3)^2}+\sqrt{(\sqrt{11}-3)^2}}$

$= \frac{\sqrt{11}}{\sqrt{11}+3+\sqrt{11}-3} = \frac{\sqrt{11}}{2\sqrt{11}} = \frac{1}{2}$

Expression $= 2+\sqrt{3}-5-\sqrt{3}+\frac{1}{2}$

$= -3+\frac{1}{2} = \frac{-6+1}{2} = \frac{-5}{2} = -2\frac{1}{2}$

18. (a); First term $= \frac{1}{\sqrt{11-2\sqrt{30}}}$

$= \sqrt{\frac{1}{11-2\sqrt{30}}} = \sqrt{\frac{1\times(11+2\sqrt{30})}{(11-2\sqrt{30})(11+2\sqrt{30})}}$

(Rationalising the denominator)

$= \sqrt{\frac{(11+2\sqrt{30})}{(11)^2-(2\sqrt{30})^2}}$

[$\because$ (a + b) (a – b) = $a^2 - b^2$]

$= \sqrt{\frac{11+2\sqrt{30}}{121-120}} = \sqrt{\frac{(11+2\sqrt{30})}{1}}$

$= \sqrt{(11+2\sqrt{30})} = \sqrt{5+6+2\times\sqrt{5}\times\sqrt{6}}$

$= \sqrt{(\sqrt{5})^2+(\sqrt{6})^2+2\sqrt{5\times6}} = \sqrt{(\sqrt{5}+\sqrt{6})^2}$

[$a^2 + b^2 + 2ab = (a + b)^2$]

$= (\sqrt{5}+\sqrt{6})$

Second term $= \frac{3}{\sqrt{7-2\sqrt{10}}}$

$= \sqrt{\frac{9}{(7-2\sqrt{10})}} = \sqrt{\frac{9\times(7+2\sqrt{10})}{(7-2\sqrt{10})(7+2\sqrt{10})}}$

$= \sqrt{\frac{9\times(7+2\sqrt{10})}{49-40}} = \sqrt{\frac{9\times(7+2\sqrt{10})}{9}}$

$= \sqrt{7+2\sqrt{10}} = \sqrt{5+2+2\sqrt{10}}$

$= \sqrt{(\sqrt{5})^2+(\sqrt{2})^2+2\sqrt{5}\times\sqrt{2}}$

$= \sqrt{(\sqrt{5}+\sqrt{2})^2} = (\sqrt{5}+\sqrt{2})$

Similarly, third term $= \frac{4}{\sqrt{8+4\sqrt{3}}}$

$= \sqrt{\frac{16}{(8+4\sqrt{3})}} = \sqrt{\frac{16\times(8-4\sqrt{3})}{(8+4\sqrt{3})(8-4\sqrt{3})}}$

$= \sqrt{\frac{16\times(8-4\sqrt{3})}{64-48}} = \sqrt{\frac{16(8-4\sqrt{3})}{16}}$

$= \sqrt{(8-4\sqrt{3})} = \sqrt{(\sqrt{6})^2+(\sqrt{2})^2-2\times\sqrt{6}\times\sqrt{2}}$

$= \sqrt{(\sqrt{6}-\sqrt{2})^2} = (\sqrt{6}-\sqrt{2})$

$\therefore$ Expression

$= \frac{1}{\sqrt{11-2\sqrt{30}}}-\frac{3}{\sqrt{7-2\sqrt{10}}}-\frac{4}{\sqrt{8+4\sqrt{3}}}$

$= (\sqrt{5}+\sqrt{6})-(\sqrt{5}+\sqrt{2})-(\sqrt{6}-\sqrt{2})$

$= \sqrt{5}+\sqrt{6}-\sqrt{5}-\sqrt{2}-\sqrt{6}+\sqrt{2} = 0$

19. (b); Expression $= \frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}-\sqrt{1-x}}$

On rationalising the denominator,

$= \frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}-\sqrt{1-x}}\times\frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}$

$= \frac{(\sqrt{1+x}+\sqrt{1-x})^2}{(\sqrt{1+x})^2-(\sqrt{1-x})^2}$

$$=\frac{1+x+1-x+2\sqrt{1+x}\cdot\sqrt{1-x}}{(1+x)-(1-x)}=\frac{1+\sqrt{1-x^2}}{x}$$

Putting $x=\frac{\sqrt{3}}{2}$

$$\therefore \text{Expression }=\frac{1+\sqrt{1-\left(\frac{\sqrt{3}}{2}\right)^2}}{\frac{\sqrt{3}}{2}}$$

$$=\frac{1+\sqrt{1-\frac{3}{4}}}{\frac{\sqrt{3}}{2}}=\frac{1+\sqrt{\frac{4-3}{4}}}{\frac{\sqrt{3}}{2}}$$

$$=\frac{1+\frac{1}{2}}{\frac{\sqrt{3}}{2}}=\frac{\frac{3}{2}}{\frac{\sqrt{3}}{2}}=\frac{3}{2}\times\frac{2}{\sqrt{3}}=\sqrt{3}$$

20. (a); $x=\frac{\sqrt{a+2b}+\sqrt{a-2b}}{\sqrt{a+2b}-\sqrt{a-2b}}$

$$=\frac{\sqrt{a+2b}+\sqrt{a-2b}}{\sqrt{a+2b}-\sqrt{a-2b}}\times\frac{\sqrt{a+2b}+\sqrt{a-2b}}{\sqrt{a+2b}+\sqrt{a-2b}}$$

$$=\frac{\left(\sqrt{a+2b}+\sqrt{a-2b}\right)^2}{\left(\sqrt{a+2b}\right)^2-\left(\sqrt{a-2b}\right)^2}$$

$$=\frac{a+2b+a-2b+2\sqrt{a+2b}\times\sqrt{a-2b}}{(a+2b)-(a-2b)}$$

$\therefore\ 2bx=a+\sqrt{a^2-4b^2}$

$\Rightarrow\ 2bx-a=\sqrt{a^2-4b^2}$

$\Rightarrow\ (2bx-a)^2=\left(\sqrt{a^2-4b^2}\right)^2$

(Squaring both sides)

$\Rightarrow\ 4b^2x^2+a^2-4abx=a^2-4b^2$

$\Rightarrow\ 4b^2x^2-4abx+4b^2=0$

$\Rightarrow\ 4b(bx^2-ax+b)=0$

$\Rightarrow\ bx^2-ax+b=0$

(Dividing both sides by 4b)

Previous Year (Memory Based)

1. (c); $\left(\frac{3}{5}\right)^3\left(\frac{3}{5}\right)^{-6}=\left(\frac{3}{5}\right)^{2x-1}$

$\Rightarrow\left(\frac{3}{5}\right)^3\left(\frac{3}{5}\right)^{-3}\left(\frac{3}{5}\right)^{-3}=\left(\frac{3}{5}\right)^{2x-1}$

$\Rightarrow\left(\frac{3}{5}\right)^0\left(\frac{3}{5}\right)^{-3}=\left(\frac{3}{5}\right)^{2x-1}$

$\Rightarrow 2x-1=-3\quad\Rightarrow\quad 2x=-3+1\quad\Rightarrow\quad x=-1$

2. (b); $\frac{\sqrt{7}-2}{\sqrt{7}+2}=\frac{\sqrt{7}-2}{\sqrt{7}+2}\times\frac{\sqrt{7}-2}{\sqrt{7}-2}$

(Rationalising the denominator)

$$=\frac{(\sqrt{7}-2)^2}{7-4}=\frac{7+4-4\sqrt{7}}{3}=\frac{11}{3}-\frac{4\sqrt{7}}{3}$$

$\therefore\ \frac{\sqrt{7}-2}{\sqrt{7}+2}=a\sqrt{7}+b\ \Rightarrow\ \frac{11}{3}-\frac{4}{3}\sqrt{7}=a\sqrt{7}+b$

Clearly, $a=-\frac{4}{3}$ and $b=\frac{11}{3}$

3. (a); $x=\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}=\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}\times\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}}$

$$=\frac{(\sqrt{5}+\sqrt{3})^2}{5-3}=\frac{5+3+2\sqrt{15}}{2}$$

$$=\frac{8+2\sqrt{15}}{2}=4+\sqrt{15}$$

$\therefore\ y=\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}=4-\sqrt{15}$

$\therefore\ x+y=4+\sqrt{15}+4-\sqrt{15}=8$

4. (b); $\sqrt{2}=1.414$ (Given)

Now, $\frac{\sqrt{2}-1}{\sqrt{2}+1}=\frac{(\sqrt{2}-1)(\sqrt{2}-1)}{(\sqrt{2}+1)(\sqrt{2}-1)}$

$$=\frac{(\sqrt{2}-1)^2}{2-1}=(\sqrt{2}-1)^2$$

$\therefore\ \sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1}}=\sqrt{(\sqrt{2}-1)^2}=\sqrt{2}-1$

$=1.414-1=0.414$

5. (b); $x+\frac{1}{4}\sqrt{x}+a^2=\left(\sqrt{x}\right)^2+2.\sqrt{x}.\frac{1}{8}+(a)^2$

Clearly $a=\frac{1}{8}$.

Then, expression $=\left(\sqrt{x}+\frac{1}{8}\right)^2$

6. **(c);** $(0.5)^2 = 0.25, \sqrt{0.49} = 0.7$

$\sqrt[3]{0.008} = 0.2, 0.23$

$\therefore \sqrt{0.49} > (0.5)^2 > 0.23 > \sqrt[3]{0.008}$

7. **(a);** $\sqrt[3]{4}, \sqrt{2}, \sqrt[6]{3}, \sqrt[4]{5}$

LCM of 3, 2, 6, 4 = 12

$\sqrt[3]{4} = (4)^{\frac{1}{3}} = (4)^{\frac{4}{12}} = (4^4)^{\frac{1}{12}} = (256)^{\frac{1}{12}}$

$\sqrt{2} = (2)^{\frac{1}{2}} = (2)^{\frac{6}{12}} = (2^6)^{\frac{1}{12}} = (64)^{\frac{1}{12}}$

$\sqrt[6]{3} = (3)^{\frac{1}{6}} = (3)^{\frac{2}{12}} = (3^2)^{\frac{1}{12}} = (9)^{\frac{1}{12}}$

$\sqrt[4]{5} = (5)^{\frac{1}{4}} = (5)^{\frac{3}{12}} = (5^3)^{\frac{1}{12}} = (125)^{\frac{1}{12}}$

$\therefore (256)^{\frac{1}{12}} > (125)^{\frac{1}{12}} > (64)^{\frac{1}{12}} > (9)^{\frac{1}{12}}$

or, $\sqrt[3]{4} > \sqrt[4]{5} > \sqrt{2} > \sqrt[6]{3}$

8. **(b);** Expression = $\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}} - \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

$= \frac{(\sqrt{3}+\sqrt{2})^2 - (\sqrt{3}-\sqrt{2})^2}{(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})}$

$= \frac{3+2+2\sqrt{6}-3-2+2\sqrt{6}}{(\sqrt{3})^2-(\sqrt{2})^2} = \frac{4\sqrt{6}}{3-2} = 4\sqrt{6}$

9. **(c);** $\sqrt{12}+\sqrt{18} = \sqrt{3\times2\times2} + \sqrt{2\times3\times3}$

$= 2\sqrt{3} + 3\sqrt{2}$

$\therefore$ Required difference

$= 2\sqrt{3}+3\sqrt{2}-2\sqrt{3}-2\sqrt{2} = \sqrt{2}$

10. **(c);** LCM of the orders of the surds = LCM of 2,3,5 and 7 = 210

$5^{\frac{1}{2}} = 5^{\frac{105}{210}} = (5^{105})^{\frac{1}{210}}$, $4^{\frac{1}{3}} = 4^{\frac{70}{210}} = (4^{70})^{\frac{1}{210}}$

$2^{\frac{1}{5}} = 2^{\frac{42}{210}} = (2^{42})^{\frac{1}{210}}$, $3^{\frac{1}{7}} = 3^{\frac{30}{210}} = (3^{30})^{\frac{1}{210}}$

$\therefore$ The largest number = $5^{\frac{1}{2}} = \sqrt{5}$

Quicker Approach

5 is the largest radicand and its order is smallest.

$\therefore$ Largest number = $\sqrt{5}$

11. **(c);** Let $x = \sqrt{7\sqrt{7\sqrt{7\sqrt{7.....}}}}$

On squaring both sides,

$x^2 = 7x \Rightarrow x^2 - 7x = 0$

$\Rightarrow x(x-7) = 0 \Rightarrow x = 7$

$\therefore 7 = (7^3)^{y-1} = 7^{3y-3}$

$\Rightarrow 3y - 3 = 1 \Rightarrow 3y = 4$

$\Rightarrow y = \frac{4}{3}$

12. **(b);** Expression = $\frac{1}{2^{\frac{2}{3}} + 2^{\frac{1}{3}} + 1}$

$= \frac{2^{\frac{1}{3}}-1}{\left(2^{\frac{1}{3}}-1\right)\left(2^{\frac{2}{3}}+2^{\frac{1}{3}}+1\right)} = \frac{2^{\frac{1}{3}}-1}{\left(2^{\frac{1}{3}}\right)^3-1}$

$= 2^{\frac{1}{3}} - 1 = \sqrt[3]{2} - 1$

[$\because$ (a – b) $(a^2 + ab + b^2) = a^3 - b^3$]

13. **(d);** $\frac{1}{\sqrt{2}+\sqrt{1}} = \frac{1}{\sqrt{2}+\sqrt{1}} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}$

$= \sqrt{2} - 1$

$\therefore$ Expression

$= \sqrt{2}-1+\sqrt{3}-\sqrt{2}+\sqrt{4}-\sqrt{3}+.....$
$+\sqrt{99}-\sqrt{98}+\sqrt{100}-\sqrt{99}$

$= \sqrt{100} - 1 = 10 - 1 = 9$

14. **(a);** $\sqrt{6}\times\sqrt{15} = x\sqrt{10}$

$\Rightarrow \sqrt{2\times3}\times\sqrt{3\times5} = x\sqrt{10}$

$\Rightarrow \sqrt{2}\times\sqrt{5}\times3 = x\sqrt{10}$

$\Rightarrow 3\sqrt{10} = x\sqrt{10} \Rightarrow x = 3$

15. **(b);** $\sqrt{7}-\sqrt{5} = \frac{(\sqrt{7}-\sqrt{5})(\sqrt{7}+\sqrt{5})}{\sqrt{7}+\sqrt{5}} = \frac{2}{\sqrt{7}+\sqrt{5}}$

Similarly,

$\sqrt{5}-\sqrt{3} = \frac{2}{\sqrt{5}+\sqrt{3}}$, $\sqrt{9}-\sqrt{7} = \frac{2}{\sqrt{9}+\sqrt{7}}$

$\sqrt{11}-\sqrt{9} = \frac{2}{\sqrt{11}+\sqrt{9}}$

$\therefore$ Largest number $= \sqrt{5}-\sqrt{3}$ because its denominator is the smallest.

16. (c); $(0.1)^2 = 0.01$

$\sqrt{0.0121} = \sqrt{0.11 \times 0.11} = 0.11$

$\sqrt{0.0004} = 0.02$, So largest number = 0.12

17. (d); Let $x = \sqrt{6+\sqrt{6+\sqrt{6+.....}}}$

On squaring both sides,

$x^2 = 6+\sqrt{6+\sqrt{6+\sqrt{6+.....}}}$

$\Rightarrow x^2 = 6 + x$

$\Rightarrow x^2 - x - 6 = 0$

$\Rightarrow x^2 - 3x + 2x - 6 = 0$

$\Rightarrow x(x-3) + 2(x-3) = 0$

$\Rightarrow (x+2)(x-3) = 0$

$\Rightarrow x = 3$ and $x \neq -2$ because numbers are positive.

18. (c); Expression $= \dfrac{(81)^{3.6} \times (9)^{2.7}}{(81)^{4.2} \times 3}$

$= \dfrac{(3^4)^{3.6} \times (3^2)^{2.7}}{(3^4)^{4.2} \times 3} = \dfrac{3^{14.4} \times 3^{5.4}}{3^{16.8} \times 3}$

$[\because (a^m)^n = a^{mn};\ a^m \times a^n = a^{m+n};\ a^m \div a^n = a^{m-n}]$

$= \dfrac{3^{14.4+5.4}}{3^{16.8+1}} = \dfrac{3^{19.8}}{3^{17.8}} = 3^{19.8-17.8} = 3^2 = 9$

19. (b); $a = \dfrac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} = \dfrac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} \times \dfrac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}}$

$= \dfrac{(\sqrt{3}-\sqrt{2})^2}{3-2} = 3+2-2\sqrt{6} = 5-2\sqrt{6}$

$\therefore\ b = \dfrac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}} = 5+2\sqrt{6}$

$\Rightarrow\ a + b = 10;$

$ab = (5-2\sqrt{6})(5+2\sqrt{6}) = 25-24 = 1$

$\therefore\ \dfrac{a^2}{b} + \dfrac{b^2}{a} = \dfrac{a^3+b^3}{ab} = \dfrac{(a+b)^3 - 3ab(a+b)}{ab}$

$= 10^3 - 3 \times 10 = 1000 - 30 = 970$

20. (d); $2\sqrt{x} = \dfrac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} - \dfrac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$

$= \dfrac{(\sqrt{5}+\sqrt{3})^2 - (\sqrt{5}-\sqrt{3})^2}{(\sqrt{5}-\sqrt{3})(\sqrt{5}+\sqrt{3})} = \dfrac{4.\sqrt{5}.\sqrt{3}}{5-3} = 2\sqrt{15}$

$\therefore\ 2\sqrt{x} = 2\sqrt{15}\ \Rightarrow\ x = 15$

21. (d); Expression

$= \dfrac{3+\sqrt{6}}{5\sqrt{3} - 2\sqrt{12} - \sqrt{32} + \sqrt{50}}$

$= \dfrac{3+\sqrt{6}}{5\sqrt{3} - 4\sqrt{3} - 4\sqrt{2} + 5\sqrt{2}}$

$= \dfrac{3+\sqrt{6}}{\sqrt{3}+\sqrt{2}} = \dfrac{\sqrt{3}(\sqrt{3}+\sqrt{2})}{\sqrt{3}+\sqrt{2}} = \sqrt{3} = 1.732$

22. (d); Expression $= \sqrt{10 + 2\sqrt{6} + 2\sqrt{10} + 2\sqrt{15}}$

$= \sqrt{10 + 2\times\sqrt{2}\times\sqrt{3} + 2\times\sqrt{2}\times\sqrt{5} + 2\times\sqrt{3}\times\sqrt{5}}$

$= \sqrt{2+3+5 + 2\times\sqrt{2}\times\sqrt{3} + 2\times\sqrt{2}\times\sqrt{5} + 2\times\sqrt{3}\times\sqrt{5}}$

$= \sqrt{(\sqrt{2})^2 + (\sqrt{3})^2 + (\sqrt{5})^2 + 2\times\sqrt{2}\times\sqrt{3} + 2\times\sqrt{2}\times\sqrt{5} + 2\times\sqrt{3}\times\sqrt{5}}$

$= \sqrt{(\sqrt{2}+\sqrt{3}+\sqrt{5})^2} = \sqrt{2}+\sqrt{3}+\sqrt{5}$

$[(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc]$

23. (c); $\sqrt{\dfrac{\sqrt{36} - \sqrt{24} + \sqrt{24} - \sqrt{16}}{5+\sqrt{24}}}$

$= \sqrt{\dfrac{6-4}{5+\sqrt{24}}} = \sqrt{\dfrac{2}{5+\sqrt{24}}} = \sqrt{\dfrac{2}{5+\sqrt{6\times4}}}$

$= \sqrt{\dfrac{2}{5+2\sqrt{6}}} = \sqrt{\dfrac{2}{5+2\sqrt{6}} \times \dfrac{5-2\sqrt{6}}{5-2\sqrt{6}}}$

$= \sqrt{\dfrac{2(5-2\sqrt{6})}{25-24}} = \sqrt{2(5-2\sqrt{6})}$

$= \sqrt{2\left[(\sqrt{3})^2 + (\sqrt{2})^2 - 2\sqrt{3}\sqrt{2}\right]}$

$= \sqrt{2(\sqrt{3}-\sqrt{2})^2} = \sqrt{2}(\sqrt{3}-\sqrt{2}) = \sqrt{6} - 2$

24. (b); $a = \dfrac{2+\sqrt{3}}{2-\sqrt{3}} = \dfrac{(2+\sqrt{3})(2+\sqrt{3})}{(2-\sqrt{3})(2+\sqrt{3})}$

$= \dfrac{4+4\sqrt{3}+3}{4-3} = 7+4\sqrt{3}$

$\therefore\ b = \dfrac{2-\sqrt{3}}{2+\sqrt{3}} = 7-4\sqrt{3}$

$\therefore a+b = 7+4\sqrt{3}+7-4\sqrt{3} = 14$

$ab = (7+4\sqrt{3})(7-4\sqrt{3})$

= 49 – 48 = 1

$\therefore a^2 + b^2 + ab = (a + b)^2 - ab$

$= (14)^2 - 1 = 196 - 1 = 195$

25. (a); Expression = $\dfrac{\sqrt{7}}{\sqrt{16+6\sqrt{7}}-\sqrt{16-6\sqrt{7}}}$

$$= \frac{\sqrt{7}}{\sqrt{9+7+2\times3\times\sqrt{7}}-\sqrt{9+7-2\times3\times\sqrt{7}}}$$

$$= \frac{\sqrt{7}}{(3+\sqrt{7})-(3-\sqrt{7})} = \frac{\sqrt{7}}{3+\sqrt{7}-3+\sqrt{7}} = \frac{1}{2}$$

26. (c); Required value will be the difference between $(\sqrt{12}+\sqrt{18})$ and $(\sqrt{3}+\sqrt{2})$

$= (\sqrt{12}+\sqrt{18}) - (\sqrt{3}+\sqrt{2})$

$= (2\sqrt{3}+3\sqrt{2}) - (\sqrt{3}+\sqrt{2})$

$= \sqrt{3}+2\sqrt{2}$

27. (d); Given, $(256)^{-(4^{-3/2})} = (256)^{-\left(\frac{1}{4^{3/2}}\right)}$

$= (256)^{-\left(\frac{1}{(2^2)^{3/2}}\right)}$

$p^{-n} = \dfrac{1}{p^n} = (256)^{-\frac{1}{8}} = \dfrac{1}{(256)^{\frac{1}{8}}} = \dfrac{1}{(2^8)^{\frac{1}{8}}} = \dfrac{1}{2}$

28. (d); $(0.9)^2 = 0.81$

$\sqrt{0.9} = 0.948, \quad 0.\overline{9} = \dfrac{9}{9} = 1$

$\therefore 0.\overline{9}$ is greatest.

29. (b); Expression = $\sqrt{\sqrt[3]{0.004096}}$

In this type of question the decimal digit is converted into decimal fraction.

$= \sqrt{\sqrt[3]{\dfrac{4096}{1000000}}}$

We know that, $\sqrt[3]{4096} = 16$

and $\sqrt[3]{1000000} = 100$

$\therefore \quad \sqrt{\dfrac{16}{100}}$

We know that, $\sqrt{16} = 4$ and $\sqrt{100} = 10$

$\therefore \quad \sqrt{\dfrac{16}{100}} = \dfrac{4}{10} = 0.4$

30. (b); $(2.89)^{0.5} = (2.89)^{\frac{1}{2}} = (1.7^2)^{1/2} = 1.7$

$\Rightarrow \; 2 - (0.5)^2 = 2 - 0.25 = 1.75,$

$\Rightarrow \; \sqrt{3} = 1.732$

and $\sqrt[3]{0.008} = \sqrt[3]{0.2\times0.2\times0.2} = 0.2$

Arrange in ascending order

$\sqrt[3]{0.008} < (2.89)^{0.5} < \sqrt{3} < 2 - (0.5)^2$